1. For what values of a does the equation $9x^2 - 2x + a = 6 - ax$ possess equal roots ?

   a के किन मानों के लिये समीकरण $9x^2 - 2x + a = 6 - ax$ के मूल समान होंगे ?

Ans. $a = 20 \pm 6\sqrt{5}$

Sol. $9x^2 - 2x + a = 6 - ax$
$9x^2 + x(a - 2) + (a - 6) = 0$
For equal roots
$D = 0$
$(a - 2)^2 - 4 \cdot 9 \cdot (a - 6) = 0$
$a^2 + 4 - 4a - 36a + 216 = 0$
$a^2 - 40a + 220 = 0$

$$a = \frac{40 \pm \sqrt{1600 - 880}}{2}$$

$$a = \frac{40 \pm \sqrt{720}}{2}$$

$$a = \frac{40 \pm 12\sqrt{5}}{2}$$

$$a = 20 \pm 6\sqrt{5}$$

2. Find the value of k for which the equation $(k - 1)x^2 + (k + 4)x + k + 7 = 0$ has equal roots.

   k के किन मानों के लिये समीकरण $(k - 1)x^2 + (k + 4)x + k + 7 = 0$ के मूल समान होंगे ?

Ans. $k_1 = -22/3$, $k_2 = 2$

Sol. $(k - 1)x^2 + (k + 4)x + k + 7 = 0$
For equal troots
$D = 0$
$(k + 4)^2 - 4(k - 1)(k + 7) = 0$
$k^2 + 16 + 8k - 4(k^2 + 6k - 7) = 0$
$k^2 + 16 + 8k - 4k^2 - 24k + 28 = 0$
$-3k^2 - 16k + 44 = 0$
$3k^2 + 16k - 44 = 0$

$$k = \frac{-16 \pm \sqrt{256 + 4 \cdot 3 \cdot 44}}{6}$$

$$k = \frac{-16 \pm \sqrt{256 + 528}}{6}$$

$$k = \frac{-16 \pm \sqrt{784}}{6}$$

$$k = \frac{-16 \pm 28}{6}$$

$$k = \frac{28 - 16}{6}, \frac{-16 - 28}{6}$$

$$k = \frac{12}{6}, -\frac{44}{6}$$

$$k = 2, \ -\frac{22}{3}$$

$$\Rightarrow k_1 = -\frac{22}{3}, \ k_2 = 2.$$

3. Find the values of a for which the roots of the equation $(2a - 5)x^2 - 2(a - 1)x + 3 = 0$ are equal.

a के किन मानों के लिये समीकरण $(2a - 5)x^2 - 2(a - 1)x + 3 = 0$ के मूल समान होंगे ?

Ans. $a = 4$

Sol. $(2a - 5)x^2 - 2(a - 1)x + 3 = 0$
For equal roots
$D = 0$
$4(a - 1)^2 - 4 \cdot 3 \cdot (2a - 5) = 0$
$a^2 + 1 - 2a - 6a + 15 = 0$
$a^2 - 8a + 16 = 0$
$(a - 4)^2 = 0$
$a = 4$

4. For what values of m does the equation $x^2 - x + m = 0$ possess no real roots ?

m के किन मानों के लिये समीकरण $x^2 - x + m = 0$ के मूल वास्तविक नहीं होंगे ?

Ans. For all $m \in (1/4, +\infty)$

Sol. $x^2 - x + m = 0$
For no real roots
$D < 0$
$1 - 4m < 0$

$$m > \frac{1}{4}$$

$m \in \left(\frac{1}{4}, +\infty\right)$

5. Form a quadratic equation whose roots are the numbers $\frac{1}{10-\sqrt{72}}$ and $\frac{1}{10+6\sqrt{2}}$.

द्विघात समीकरण ज्ञात कीजिये जिसके मूल $\frac{1}{10-\sqrt{72}}$ तथा $\frac{1}{10+6\sqrt{2}}$ हैं।

Ans. $28x^2 - 20x + 1 = 0$

Sol. Let $\alpha = \frac{1}{10-\sqrt{72}} = \frac{1}{10-6\sqrt{2}} = \frac{10+6\sqrt{2}}{100-72} = \frac{1}{28}\left(10+6\sqrt{2}\right)$

and $\beta = \frac{1}{10+6\sqrt{2}} = \frac{10-6\sqrt{2}}{100-72} = \frac{1}{28}\left(10-6\sqrt{2}\right)$

Sum of roots = $\alpha+\beta = \frac{10+6\sqrt{2}+10-6\sqrt{2}}{28} = \frac{20}{28} = \frac{5}{7}$

Product of roots = $\alpha\cdot\beta = \frac{\left(10+6\sqrt{2}\right)\cdot\left(10-6\sqrt{2}\right)}{28\cdot 28} = \frac{28}{28\cdot 28} = \frac{1}{28}$

Then Q.E.

$x^2 - (\alpha + \beta)\, x + \alpha\beta = 0$

$x^2 - \frac{5}{7}x + \frac{1}{28} = 0$

$28x^2 - 20x + 1 = 0$

6. Find the least integral value of k for which the equation $x^2 - 2(k+2)x + 12 + k^2 = 0$ has two different real roots.

k के न्यूनतम पूर्णांक मान ज्ञात कीजिये जिसके लिये समीकरण $x^2 - 2(k+2)x + 12 + k^2 = 0$ के दो भिन्न वास्तविक मूल होंगे।

Ans. $k = 3$

Sol. $x^2 - 2(k+2)x + 12 + k^2 = 0$

For two distinct real roots

$D > 0$

$4(k+2)^2 - 4\cdot(12 + k^2) > 0$

$k^2 + 4 + 4k - 12 - k^2 > 0$

$4k - 8 > 0$

$k > 2$

Least integral value of $k = 3$.

7. For what values of a is the sum of the roots of the equation $x^2 + (2 - a - a^2)x - a^2 = 0$ equal to zero ?

a का मान ज्ञात कीजिये जिसके लिये समीकरण $x^2 + (2 - a - a^2)x - a^2 = 0$ के मूलों का योग शून्य है ?

Ans. $a_1 = -2, a_2 = 1$

Sol. $x^2 + (2 - a - a^2)x - a^2 = 0$
Let roots are α and β.
$\because$ sum of roots = 0
$\alpha + \beta = 0$
$2 - a - a^2 = 0$
$a^2 + a - 2 = 0$
$(a - 1)(a + 2) = 0$
$a = 1, -2$

8. Find the value of a for which one root of the equation $x^2 + (2a - 1)x + a^2 + 2 = 0$ is twice as large as the other.

a का मान ज्ञात कीजिये जिसके लिये समीकरण $x^2 + (2a - 1)x + a^2 + 2 = 0$ का एक मूल दूसरे का दुगुना है ?

Ans. $a = -4$

Sol. $a^2 + (2a - 1)x + a^2 + 2 = 0$
roots : $\alpha, 2\alpha$
sum of roots = $\alpha + 2\alpha = -(2a - 1)$
$3\alpha = (1 - 2a)$

$$\alpha = \frac{(1-2a)}{3}$$

Poduct of roots = $\alpha \cdot 2\alpha = (a^2 + 2)$
$2\alpha^2 = a^2 + 2$

$$2 \cdot \frac{(1-2a)^2}{9} = a^2 + 2$$

$2(1 + 4a^2 - 4a) = 9(a^2 + 2)$
$2 + 8a^2 - 8a = 9a^2 + 18$
$a^2 + 8a + 16 = 0$
$(a + 4)^2 = 0$
$a = -4$

9. For what values of a is the ratio of the roots of the equation $x^2 + ax + a + 2 = 0$ equal to 2 ?

a का मान ज्ञात कीजिये जिसके लिये समीकरण $x^2 + ax + a + 2 = 0$ के मूलों का अनुपात 2 है ?

Ans. $a_1 = -3/2, a_2 = 6$

Sol. $x^2 + ax + a + 2 = 0$
α, β are the roots

$\because \frac{\alpha}{\beta} = 2$

$\alpha = 2\beta$

Sum of roots = –a

$\alpha + \beta = -a$

$3\beta = -a$

$\beta = -\frac{a}{3}$

Product of roots = (a + 2)

$\alpha \cdot \beta = (a + 2)$

$2\beta \cdot \beta = (a + 2)$

$2\left(-\frac{a}{3}\right)^2 = (a+2)$

$2a^2 = 9(a + 2)$

$2a^2 - 9a - 18 = 0$

$2a^2 - 12a + 3a - 18 = 0$

$2a(a - 6) + 3(a - 6) = 0$

$(a - 6)(2a + 3) = 0$

$a = -\frac{3}{2}, 6$

10. For what values of a is the ratio of the roots of the equation $ax^2 - (a + 3)x + 3 = 0$ equal to 1.5 ?

a का मान ज्ञात कीजिये जिसके लिये समीकरण $ax^2 - (a + 3)x + 3 = 0$ के मूलों का अनुपात 1.5 है ?

Ans. $a_1 = 2, a_2 = 9/2$

Sol. $ax^2 - (a + 3)x + 3 = 0$

α, β are the roots

$\because \frac{\alpha}{\beta} = 1.5 = \frac{3}{2}$

$\alpha = \frac{3}{2}\beta$

$\Rightarrow \alpha + \beta = \frac{(a+3)}{a}$

$\frac{3}{2}\beta + \beta = \frac{(a+3)}{a}$

$\frac{5}{2}\beta = \frac{(a+3)}{a}$

$$\beta = \frac{2(a+3)}{5a}$$

$$\Rightarrow \alpha\cdot\beta = \frac{3}{a}$$

$$\frac{3}{2}\beta\cdot\beta = \frac{3}{a}$$

$$\frac{1}{2}\beta^2 = \frac{1}{a}$$

$$\frac{1}{2}\cdot\frac{4(a+3)^2}{25a^2} = \frac{1}{a}$$

$2(a^2 + 9 + 6a) = 25a$
$2a^2 + 12a + 18 - 25a = 0$
$2a^2 - 13a + 18 = 0$
$2a^2 - 4a - 9a + 18 = 0$
$2a(a - 2) - 9(a - 2) = 0$
$(a - 2)(2a - 9) = 0$

$$a = 2, \frac{9}{2}$$

11. For what values of a do the roots x_1 and x_2 of the equation $x^2 - (3a + 2)x + a^2 = 0$ satisfy the relation $x_1 = 9x_2$? Find the roots.

यदि समीकरण $x^2 - (3a + 2)x + a^2 = 0$ के मूल x_1 तथा x_2 सम्बन्ध $x_1 = 9x_2$ को सन्तुष्ट करते है तो मूल ज्ञात कीजिये ?

Ans. {2, 18} for a = 6, {2/19, 18/19} for a = –6/19

Sol. $x^2 - (3a + 2)x + a^2 = 0$
x_1, x_2 are the roots
$\because\ x_1 = 9x_2$
$\Rightarrow x_1 + x_2 = 3a + 2$
$10x_2 = 3a + 2$

$$x_2 = \frac{3a+2}{10}$$

$\Rightarrow x_1\cdot x_2 = a^2$
$9x_2^2 = a^2$

$$9\cdot\frac{(3a+2)^2}{100} = a^2$$

$9(9a^2 + 4 + 12a) = 100a^2$
$100a^2 - 81a^2 - 108a - 36 = 0$
$19a^2 - 108a - 36 = 0$

$$a = \frac{108 \pm \sqrt{(108)^2 + 4 \cdot 19 \cdot 36}}{2 \times 19}$$

$$a = \frac{108 \pm \sqrt{11664 + 2736}}{2 \times 19}$$

$$a = \frac{108 \pm \sqrt{14400}}{2 \times 19}$$

$$a = \frac{108 \pm 120}{2 \times 19}$$

$$a = \frac{228}{2 \times 19}, -\frac{12}{2 \times 19}$$

$$a = 6, -\frac{6}{19}$$

For $a = 6$

$$x_2 = \frac{3 \times 6 + 2}{10} = \frac{20}{10} = 2$$

$$x_1 = 9 \times 2 = 18$$

For $a = -\frac{6}{19}$

$$x_2 = \frac{3 \times \left(-\frac{6}{19}\right) + 2}{10} = \frac{-18 + 38}{10 \times 19} = \frac{20}{10 \times 19} = \frac{2}{19}$$

$$x_1 = 9 \times \frac{2}{19} = \frac{18}{19}$$

12. Find a such that one of the roots of the equation $x^2 - (15/4)x + a = 0$ is the square of the other.

a का मान ज्ञात कीजिये जिसके लिये समीकरण $x^2 - (15/4)x + a = 0$ का एक मूल दूसरे का वर्ग है ?

Ans. $a_1 = -125/8$, $a_2 = 27/8$

Sol. $$x^2 - \left(\frac{15}{4}\right)x + a = 0$$

roots = α, α^2

$$\text{Sum of roots} = \frac{15}{4}$$

$$\alpha + \alpha^2 = \frac{15}{4}$$

$4\alpha^2 + 4\alpha - 15 = 0$ ...(i)
$4\alpha^2 + 10\alpha - 6\alpha - 15 = 0$
$2\alpha\cdot(2\alpha + 5) - 3(2\alpha + 5) = 0$
$(2\alpha + 5)(2\alpha - 3) = 0$

$$\alpha = -\frac{5}{2}, \frac{3}{2}$$

Product of roots = a
$\alpha^3 = a$

for $\alpha = -\frac{5}{2}$

$$a = \left(-\frac{5}{2}\right)^3 = \frac{125}{8}$$

for $\alpha = \frac{3}{2}$

$$a = \left(\frac{3}{2}\right)^3 = \frac{27}{8}$$

13. The roots x_1 and x_2 of the equation $x^2 + px + 12 = 0$ are such that $x_2 - x_1 = 1$. Find p.

यदि समीकरण $x^2 + px + 12 = 0$ के मूल x_1 तथा x_2 सम्बन्ध $x_2 - x_1 = 1$ को सन्तुष्ट करते है तो p ज्ञात कीजिये ?

Ans. $p = \pm 7$

Sol. $x^2 + px + 12 = 0$
x_1, x_2 are the roots
$\Rightarrow x_1 + x_2 = -p$
$\Rightarrow x_1\cdot x_2 = 12$
$\because\ x_2 - x_1 = 1$ (Given)
$\Rightarrow (x_1 + x_2)^2 = (x_1 - x_2)^2 + 4x_1x_2$
$(-p)^2 = (1)^2 + 4\cdot 12$
$p^2 = 1 + 48$
$p^2 = 49$
$p = \pm 7$

14. Find k in the equation $5x^2 - kx + 1 = 0$ such that the difference between the roots of the equation is unity.

समीकरण $5x^2 - kx + 1 = 0$ में k का मान ज्ञात कीजिये यदि मूलों का अन्तर इकाई हो।

Ans. $k = \pm 3\sqrt{5}$

Sol. $5x^2 - kx + 1 = 0$
x_1, x_2 are the roots

$$x_1 + x_2 = \frac{k}{5}$$

$$x_1 \cdot x_2 = \frac{1}{5}$$

Given $|x_1 - x_2| = 1$

$\because\ (x_1 + x_2)^2 = (x_1 - x_2)^2 + 4x_1x_2$

$$\frac{k^2}{25} = 1 + \frac{4}{5}$$

$$\frac{k^2}{25} = \frac{9}{5}$$

$k^2 = 45$

$k = \pm 3\sqrt{5}$

15. For what value of a is the difference between the roots of the equation $(a-2)x^2 - (a-4)x - 2 = 0$ equal to 3 ?

समीकरण $(a-2)x^2 - (a-4)x - 2 = 0$ में a का मान ज्ञात कीजिये यदि मूलों का अन्तर 3 हो ?

Ans. $a_1 = 3/2,\ a_2 = 3$

Sol. $(a-2)x^2 - (a-4)x - 2 = 0$

a_1, a_2 are the roots

$$a_1 + a_2 = \frac{(a-4)}{(a-2)}$$

$$a_1 \cdot a_2 = \frac{-2}{(a-2)}$$

Given $|a_1 - a_2| = 3$

$\because\ (a_1 + a_2)^2 = (a_1 - a_2)^2 + 4a_1 \cdot a_2$

$$\frac{(a-4)^2}{(a-2)^2} = 9 + 4\cdot\left(\frac{-2}{(a-2)}\right)$$

$$\frac{(a-4)^2}{(a-2)^2} = \frac{9(a-2)-8}{(a-2)}$$

$(a-4)^2 = 9(a-2)^2 - 8(a-2)$

$a^2 + 16 - 8a = 9\cdot(a^2 + 4 - 4a) - 8\,(a-2)$

$a^2 + 16 - 8a = 9a^2 + 36 - 36a - 8a + 16$

$8a^2 - 36a + 36 = 0$

$2a^2 - 9a + 9 = 0$

$2a^2 - 6a - 3a + 9 = 0$

$2a(a-3) - 3(a-3) = 0$

$(a-3)(2a-3)=0$

$a=3,\frac{3}{2}$

16. Find b in the equation $5x^2+bx-28=0$ if the roots x_1 and x_2 of the equation are related as $5x_1+2x_2=1$ and b is an integer.

यदि समीकरण $5x^2+bx-28=0$ के मूल x_1 तथा x_2 सम्बन्ध $5x_1+2x_2=1$ को सन्तुष्ट करते है तो b ज्ञात कीजिये (b एक पूर्णांक है) ?

Ans. $b=-13$

Sol. $5x^2+bx-28=0$

x_1, x_2 are roots

$x_1+x_2=-\frac{b}{5}$

$x_1 \cdot x_2=-\frac{28}{5}$

& $5x_1+2x_2=1$

$x_1=\frac{(1-2x_2)}{5}$

$\Rightarrow \frac{(1-2x_2)}{5}\cdot x_2=-\frac{28}{5}$

$-2x_2^2+x_2+28=0$

$2x_2^2-x_2-28=0$

$2x_2^2-8x_2+7x_2-28=0$

$2x_2(x_2-4)+7(x_2-4)=0$

$(x_2-4)(2x_2+7)=0$

$x_2=4,-\frac{7}{2}$

$x_1=\frac{1-8}{5},\frac{1+2\times\frac{7}{2}}{5}$

$=-\frac{7}{5},\frac{8}{5}$

For $x_1=-\frac{7}{5},\ x_2=4$

$\Rightarrow -\frac{7}{5}+4=-\frac{b}{5}$

$b = -13$

For $x_1 = \frac{8}{5}, x_2 = -\frac{7}{2}$

$\frac{8}{5} - \frac{7}{2} = -\frac{b}{5}$

$\frac{16-35}{5\times 2} = -\frac{b}{5}$

$b = \frac{19}{2} \notin I$

$\therefore\ b = -13$

17. Find p in the equation $x^2 - 4x + p = 0$ if it is known that the sum of the squares of its roots is equal to 16.

समीकरण $x^2 - 4x + p = 0$ में p का मान ज्ञात कीजिये यदि मूलों के वर्गों का योग 16 है।

Ans. $p = 0$

Sol. $x^2 - 4x + p = 0$
α, β are the roots
$\alpha + \beta = 4$
$\alpha\beta = p$
$\therefore\ \alpha^2 + \beta^2 = 16$
$(\alpha + \beta)^2 - 2\alpha\beta = 16$
$16 - 2p = 16$
$p = 0$

18. For what values of a is the difference between the roots of the equation $2x^2 - (a + 1)x + (a - 1) = 0$ equal to their product ?

समीकरण $2x^2 - (a + 1)x + (a - 1) = 0$ में a का मान ज्ञात कीजिये यदि मूलों का अन्तर उनके गुणन के बराबर है।

Ans. $a = 2$

Sol. $2x^2 - (a + 1)x + (a - 1) = 0$
α, β are the roots.

$\alpha + \beta = \frac{(a+1)}{2}$

$\alpha\cdot\beta = \frac{(a-1)}{2}$

$|\alpha - \beta| = \left|\frac{\sqrt{D}}{a}\right|$

$$|\alpha-\beta|=\left|\frac{\sqrt{(a+1)^2-8(a-1)}}{2}\right|$$

$$\alpha\cdot\beta=\left|\frac{\sqrt{a^2-6a+9}}{2}\right|$$

$$\frac{(a-1)}{2}=\left|\frac{(a-3)}{2}\right|$$

$$\frac{(a-3)}{2}=\pm\frac{(a-1)}{2}$$

For positive
a – 3 = a – 1 (Impossible)
For negative
(a – 3) = – (a – 1)
a – 3 = –a + 1
2a = 4
a = 2

19. Find all values of a for which the sum of the roots of the equation $x^2 - 2a\,(x-1) - 1 = 0$ is equal to the sum of the squares of its roots.

समीकरण $x^2 - 2a\,(x-1) - 1 = 0$ में a का मान ज्ञात कीजिये यदि मूलों का योग उनके वर्गों के योग के बराबर है।

Ans. $a_1 = 1/2,\ a_2 = 1$

Sol. $x^2 - 2a\,(x-1) - 1 = 0$
$x^2 - 2ax + (2a-1) = 0$
α, β are the roots.
$\alpha + \beta = 2a$
$\alpha\cdot\beta = (2a-1)$
$\because\ \alpha + \beta = \alpha^2 + \beta^2$
$2a = (\alpha+\beta)^2 - 2\alpha\beta$
$2a = 4a^2 - 2\,(2a-1)$
$4a^2 - 4a + 2 - 2a = 0$
$4a^2 - 6a + 2 = 0$
$2a^2 - 3a + 1 = 0$
$2a^2 - 2a - a + 1 = 0$
$2a\,(a-1) - 1\,(a-1) = 0$
$(a-1)\,(2a-1) = 0$

$$a = 1, \frac{1}{2}$$

20. Find the coefficients of the equation $x^2 + px + q = 0$ such that its roots are equal to p and q.

समीकरण $x^2 + px + q = 0$ के गुणांक ज्ञात कीजिये यदि मूल p तथा q है।

Ans. $p_1 = 0, q_1 = 0; p_2 = 1, q_2 = -2$

Sol. $x^2 + px + q = 0$
p,q are the roots.
$p + q = -p$
$q = -2p$ ...(1)
and $pq = q$
$q(p-1) = 0$
$q = 0$ or $p = 1$
at $q = 0$
$p = 0$
at $p = 1$
$q = -2$

21. Express $x^3_1 + x^3_2$ in the terms of the coefficients of the equation $x^2 + px + q = 0$, where x_1 and x_2 are the roots of the equation.

$x^3_1 + x^3_2$ को समीकरण $x^2 + px + q = 0$ के गुणांकों के पदों में व्यक्त कीजिये जहाँ समीकरण के मूल x_1 तथा x_2 है।

Ans. $x^3_1 + x^3_2 = 3pq - p^3$

Sol. $x^2 + px + q = 0$
x_1, x_2 are the roots.
$x_1 + x_2 = -p$
$x_1 \cdot x_2 = q$
$\Rightarrow x_1^3 + x_2^3 = (x_1 + x_2)^3 - 3x_1x_2(x_1 + x_2)$
$= -p^3 - 3q(-p)$
$= 3pq - p^3$

22. Assume that x_1 and x_2 are roots of the equation $3x^2 - ax + 2a - 1 = 0$. Calculate $x^3_1 + x^3_2$.

समीकरण $3x^2 - ax + 2a - 1 = 0$ के मूल x_1 तथा x_2 है तो $x^3_1 + x^3_2$ ज्ञात कीजिये।

Ans. $x^3_1 + x^3_2 = a(a^2 - 18a + 9)/27$

Sol. $3x^2 - ax + 2a - 1 = 0$
x_1, x_2 are the roots

$$x_1 + x_2 = \frac{a}{3}$$

$$x_1 \cdot x_2 = \frac{(2a-1)}{3}$$

$$x_1^3 + x_2^3 = (x_1 + x_2)^3 - 3x_1x_2(x_1 + x_2)$$

$$= \left(\frac{a}{3}\right)^3 - 3\cdot\frac{(2a-1)}{3}\cdot\frac{a}{3}$$

$$= \frac{a^3}{27} - \frac{a(2a-1)}{3}$$

$$= \frac{a\left(a^2 - 18a + 9\right)}{27}$$

23. Without solving the equation $3x^2 - 5x - 2 = 0$, find the sum of the cubes of its roots.

समीकरण $3x^2 - 5x - 2 = 0$ को हल किये बिना मूलों के घनों का योग ज्ञात कीजिये।

Ans. 215/27

Sol. $3x^2 - 5x - 2 = 0$

x_1, x_2 are the roots.

$$x_1 + x_2 = \frac{5}{3}$$

$$x_1 \cdot x_2 = -\frac{2}{3}$$

$$\Rightarrow x_1^3 + x_2^3 = (x_1 + x_2)^3 - 3x_1x_2(x_1 + x_2)$$

$$= \left(\frac{5}{3}\right)^3 - 3\cdot\left(-\frac{2}{3}\right)\cdot\left(\frac{5}{3}\right)$$

$$= \frac{125}{27} + \frac{10}{3}$$

$$= \frac{125 + 90}{27}$$

$$= \frac{215}{27}$$

24. Calculate $\frac{1}{x_1^3} + \frac{1}{x_2^3}$, where x_1 and x_2 are roots of the equation $2x^2 - 3ax - 2 = 0$.

समीकरण $2x^2 - 3ax - 2 = 0$ के मूल x_1 तथा x_2 है तो $\frac{1}{x_1^3} + \frac{1}{x_2^3}$ ज्ञात कीजिये।

Ans. $\frac{1}{x_1^3}+\frac{1}{x_2^3}=-\frac{27a^3+36a}{8}$

Sol. $2x^2-3ax-2=0$

x_1, x_2 are the roots.

$$x_1+x_2=\frac{3a}{2}$$

$$x_1\cdot x_2=-1$$

$$x_1^3+x_2^3=(x_1+x_2)^3-3x_1x_2(x_1+x_2)$$

$$=\left(\frac{3a}{2}\right)^3-3(-1)\cdot\left(\frac{3a}{2}\right)$$

$$=\frac{27a^3}{8}+\frac{9a}{2}$$

$$=\frac{27a^3+36a}{8}$$

$$\Rightarrow \frac{1}{x_1^3}+\frac{1}{x_2^3}=\frac{x_1^3+x_2^3}{(x_1\cdot x_2)^3}=\frac{27a^3+36a}{8\times(-1)}=-\frac{27a^3+36a}{8}$$

1. Form an equation whose roots are $-\frac{4}{5}$ and $\frac{3}{7}$.

   वह समीकरण ज्ञात कीजिए जिसके मूल $-\frac{4}{5}$ तथा $\frac{3}{7}$ है।

Ans. $35x^2 + 13x - 12 = 0$

Sol. Sum of roots $= -\frac{4}{5}+\frac{3}{7}=\frac{-28+15}{35}=\frac{-13}{35}$

Product of roots $= \frac{-4}{5}\times\frac{3}{7}=-\frac{12}{35}$

Now Quadratic Equation $x^2-(\alpha+\beta)x+\alpha\beta=0$

$x^2+\frac{13}{35}x-\frac{12}{35}$

$35x^2 + 13x - 12 = 0$

2. Form an equation whose roots are $\frac{m}{n},\frac{-n}{m}$.

   वह समीकरण ज्ञात कीजिए जिसके मूल $\frac{m}{n},\frac{-n}{m}$ है।

Ans. $mnx^2 + x(n^2 - m^2) - mn = 0$

Sol. Sum of roots $= \frac{m}{n}-\frac{n}{m}=\frac{m^2-n^2}{mn}$

Product of roots $= \frac{m}{n}\times\left(-\frac{n}{m}\right)=-1$

Now Quadratic Equation $x^2 - (\alpha + \beta)x + \alpha\beta = 0$

$x^2-\frac{\left(m^2-n^2\right)}{mn}x-1=0$

$mnx^2 + (n^2 - m^2)x - mn = 0$

3. Find the equation whose roots are $\frac{p-q}{p+q}, -\frac{p+q}{p-q}$

वह समीकरण ज्ञात कीजिए जिसके मूल $\frac{p-q}{p+q}, -\frac{p+q}{p-q}$ है।

Ans. $x^2(p^2-q^2)+4pqx+(q^2-p^2)=0$

Sol. Sum of roots $= \frac{p-q}{p+q}-\frac{p+q}{p-q}=\frac{(p-q)^2-(p+q)^2}{p^2-q^2}=\frac{p^2+q^2-2pq-p^2-q^2-2pq}{p^2-q^2}=\frac{-4pq}{p^2-q^2}$

Product of roots $= -\left(\frac{p-q}{p+q}\right)\cdot\left(\frac{p+q}{p-q}\right)=-1$

Now Quadratic Equation $x^2-(\alpha+\beta)x+\alpha\beta=0$

$x^2+\frac{4pq}{(p^2-q^2)}x-1=0$

$(p^2-q^2)x^2+4pqx+(q^2-p^2)=0$

4. Find the equation whose roots are $7\pm2\sqrt{5}$

वह समीकरण ज्ञात कीजिए जिसके मूल $7\pm2\sqrt{5}$ है।

Ans. $x^2-14x+29=0$

Sol. Sum of roots $= 7+2\sqrt{5}+7-2\sqrt{5}=14$

Product of roots $= \left(7+2\sqrt{5}\right)\cdot\left(7-2\sqrt{5}\right)$

$= 49-20=29$

Now Quadratic Equation $x^2-(\alpha+\beta)x+\alpha\beta=0$

$x^2-14x+29=0$

5. Find the equation whose roots are $\pm2\sqrt{3}-5$

वह समीकरण ज्ञात कीजिए जिसके मूल $\pm2\sqrt{3}-5$ है।

Ans. $x^2+10x+13=0$

Sol. Sum of roots $= 2\sqrt{3}-5-2\sqrt{3}-5=-10$

Product of roots $= -\left(2\sqrt{3}-5\right)\left(2\sqrt{3}+5\right)=-(12-25)=13$

Now Quadratic Equation $x^2 - (\alpha + \beta)x + \alpha\beta = 0$
$x^2 + 10x + 13 = 0$

6. Find the equation whose roots are $-p \pm 2\sqrt{2q}$

वह समीकरण ज्ञात कीजिए जिसके मूल $-p \pm 2\sqrt{2q}$ है।

Ans. $x^2 + 2px + (p^2 - 8q) = 0$

Sol. Sum of roots $= -p + 2\sqrt{2q} - p - 2\sqrt{2q} = -2p$

Product of roots $= -\left(-p + 2\sqrt{2q}\right)\left(p + 2\sqrt{2q}\right) = -\left(8q - p^2\right)$
Now Quadratic Equation $x^2 - (\alpha + \beta)x + \alpha\beta = 0$
$x^2 + 2px + (p^2 - 8q) = 0$

7. Find the equation whose roots are $-3 \pm 5i$

वह समीकरण ज्ञात कीजिए जिसके मूल $-3 \pm 5i$ है।

Ans. $x^2 + 6x + 34 = 0$

Sol. Sum of roots $= -3 + 5i - 3 - 5i = -6$
Product of roots $= (-3 + 5i)(-3 - 5i)$
$= -(-3 + 5i)(3 + 5i)$
$= -(-25 - 9) = 34$
Now Quadratic Equation $x - (\alpha + \beta)x + \alpha\beta = 0$
$x^2 + 6x + 34 = 0$

8. Find the equation whose roots are $-a \pm ib$

वह समीकरण ज्ञात कीजिए जिसके मूल $-a \pm ib$ है।

Ans. $x^2 + 2ax + a^2 + b^2 = 0$

Sol. Sum of roots $= -a + ib - a - ib = -2a$
Product of roots $= (-a + ib)(-a - ib)$
$= -(-a + ib)(a + ib)$
$= -(-a^2 - b^2) = a^2 + b^2$
Now Quadratic Equation $x^2 - (\alpha + \beta)x + \alpha\beta = 0$
$x^2 + 2ax + (a^2 + b^2) = 0$

9. Find the equation whose roots are $\pm i(a - b)$

वह समीकरण ज्ञात कीजिए जिसके मूल $\pm i\,(a-b)$ है।

Ans. $x^2 + a^2 - 2ab + b^2 = 0$

Sol. Sum of roots = $i(a-b) - i(a-b) = i(a-b-a+b) = 0$
Product of roots = $-i(a-b) \times i(a-b) = +(a-b)^2$
Now Quadratic Equation = $x^2 - (\alpha+\beta)x + \alpha\beta = 0$
$x^2 - 0x + (a-b)^2 = 0$
$x^2 + (a-b)^2 = 0$
$x^2 + a^2 - 2ab + b^2 = 0$

10. Find the equation whose roots are $-3, \frac{2}{3}, \frac{1}{2}$

वह समीकरण ज्ञात कीजिए जिसके मूल $-3, \frac{2}{3}, \frac{1}{2}$ है।

Ans. $6x^3 + 11x^2 - 19x + 6$

Sol. Roots = $-3, \frac{2}{3}, \frac{1}{2}$

Equation : $(x+3)\left(x-\frac{2}{3}\right)\left(x-\frac{1}{2}\right) = 0$

$(x+3)(3x-2)(2x-1) = 0$
$(3x^2 - 2x + 9x - 6)(2x-1) = 0$
$(3x^2 + 7x - 6)(2x-1) = 0$
$6x^3 - 3x^2 + 14x^2 - 7x - 12x + 6 = 0$
$6x^3 + 11x^2 - 19x + 6 = 0$

11. Find the equation whose roots are $\frac{a}{2}, 0, -\frac{2}{a}$

वह समीकरण ज्ञात कीजिए जिसके मूल $\frac{a}{2}, 0, -\frac{2}{a}$ है।

Ans. $2ax^3 + (4 - a^2)x^2 - 2ax = 0$

Sol. Roots = $\frac{a}{2}, 0, -\frac{2}{a}$

Equation : $\left(x-\frac{a}{2}\right)\left(x+\frac{2}{a}\right)(x-0) = 0$

$$x\cdot\frac{(2x-a)}{2}\cdot\frac{(ax+2)}{a}=0$$

$x\cdot(2ax^2+4x-a^2x-2a)=0$

$2ax^3+(4-a^2)x^2-2ax=0$

12. Find the equation whose roots are $2\pm\sqrt{3}, 4$

वह समीकरण ज्ञात करो जिसके मूल $2\pm\sqrt{3}, 4$ है।

Ans. $x^3-8x^2+17x-4=0$

Sol. Roots = $2+\sqrt{3}, 2-\sqrt{3}, 4$

Equation : $\left(x-\left(2+\sqrt{3}\right)\right)\left(x-\left(2-\sqrt{3}\right)\right)(x-4)=0$

$\left((x-2)-\sqrt{3}\right)\cdot\left((x-2)+\sqrt{3}\right)(x-4)=0$

$\left\{(x-2)^2-3\right\}(x-4)=0$

$\left\{x^2-4x+4-3\right\}(x-4)=0$

$x^3-4x^2-4x^2+16x+x-4=0$

$x^3-8x^2+17x-4=0$

13. Prove that the roots of the following equation are real $x^2-2ax+a^2-b^2-c^2=0$

सिद्ध करो कि समीकरण $x^2-2ax+a^2-b^2-c^2=0$ के मूल वास्तविक है।

Ans.

Sol. Quadratic Equation : $x^2-2ax+a^2-b^2-c^2=0$

Now $D=(-2a)^2-4(a^2-b^2-c^2)$

$=4a^2-4a^2+b^2+c^2$

$=b^2+c^2>0$

$\therefore$ Roots are real. Hence proved.

14. If the equation $x^2-2mx+8m-15=0$ has equal roots, find the value of m.

यदि समीकरण $x^2-2mx+8m-15=0$ के मूल बराबर हो तो m का मान ज्ञात कीजिए।

Ans. $m=3, 5$

Sol. If equation $x^2-2mx+8m-15=0$ has equal roots.

$\therefore D=0$

$(-2m)^2-4(8m-15)=0$

$4m^2 - 32m + 60 = 0$
$m^2 - 8m + 15 = 0$
$m^2 - 5m - 3m + 15 = 0$
$m(m-5) - 3(m-5) = 0$
$(m-5)(m-3) = 0$
$m = 5, 3.$

15. For what value of m will the equation $x^2 - 2x(1 + 3m) + 7(3 + 2m) = 0$ have equal roots?

m के किस मान के लिए समीकरण $x^2 - 2x(1 + 3m) + 7(3 + 2m) = 0$ के मूल बराबर है ?

Ans. $m = 2, \frac{-10}{9}$

Sol. If equation $x^2 - 2x(1 + 3m) + 7(3 + 2m) = 0$ has equal roots.
$\therefore D = 0$
$(-2(1 + 3m))^2 - 4 \times 7(3 + 2m) = 0$
$4(1 + 3m)^2 - 4 \times (21 + 14m) = 0$
$1 + 9m^2 + 6m - 21 - 14m = 0$
$9m^2 - 8m - 20 = 0$
$9m(m-2) + 10(m-2) = 0$
$(m-2)(9m + 10) = 0$

$$m = 2, -\frac{10}{9}$$

16. For what value of m, will the equation $\frac{x^2 - bx}{ax - c} = \frac{m-1}{m+1}$ have roots equal in magnitude but opposite in sign ?

m के किस मान के लिए समीकरण $\frac{x^2 - bx}{ax - c} = \frac{m-1}{m+1}$ के मूल बराबर तथा विपरीत चिन्ह के होंगे।

Ans. $m = \frac{a-b}{a+b}$

Sol. Equation : $\frac{x^2 - bx}{ax - c} = \frac{m-1}{m+1}$
$(m + 1)x^2 - b(m + 1)x = a(m-1)x - c(m-1)$
$(m + 1)x^2 - \{b(m + 1) + a(m-1)\}x + c(m-1) = 0$
$\because$ roots of the equation are equal in magnitude but opposite in sign.
$\therefore$ Let roots = $\alpha, -\alpha$

Now $\alpha + (-\alpha) = \frac{b(m+1) + a(m-1)}{(m+1)}$

$bm + b + am - a = 0$
$m \cdot (a + b) + (b - a) = 0$

$$m = \frac{(a-b)}{(a+b)}$$

17. If α, β are the roots of the equation $ax^2 + bx + c = 0$ find the value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$.

यदि α, β समीकरण $ax^2 + bx + c = 0$ के मूल हो तो $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ का मान ज्ञात कीजिए।

Ans. $\frac{b^2 - 2ac}{c^2}$

Sol. Equation $ax^2 + bx + c = 0$
roots are α, β

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha \cdot \beta = \frac{c}{a}$$

Now $$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{\alpha^2\beta^2} = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{(\alpha\beta)^2} = \frac{\frac{b^2}{a^2} - \frac{2c}{a}}{\frac{c^2}{a^2}}$$

$$= \frac{b^2 - 2ac}{c^2}$$

18. If α, β are the roots of the equation $ax^2 + bx + c = 0$, find the value of $\alpha^4\beta^7 + \alpha^7\beta^4$

यदि α, β समीकरण $ax^2 + bx + c = 0$ के मूल हो तो $\alpha^4\beta^7 + \alpha^7\beta^4$ का मान ज्ञात कीजिए।

Ans. $\frac{c^4(3abc - b^3)}{a^7}$

Sol. Equation $ax^2 + bx + c = 0$
α, β are the roots.

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha \cdot \beta = \frac{c}{a}$$

Now $\alpha^4\beta^7 + \alpha^7\beta^4$

$= \alpha^4\beta^4 (\alpha^3 + \beta^3)$

$= \alpha^4\beta^4 \{(\alpha + \beta)^3 - 3 (\alpha + \beta) \alpha\beta\}$

$$= \frac{c^4}{a^4}\left\{-\frac{b^3}{a^3} + \frac{3b}{a}\cdot\frac{c}{a}\right\}$$

$$= \frac{c^4}{a^4}\cdot\left\{\frac{-b^3 + 3abc}{a^3}\right\}$$

$$= \frac{c^4\left(3abc - b^3\right)}{a^7}$$

19. If α, β are the roots of the equation $ax^2 + bx + c = 0$, find the value of $\left(\frac{\alpha}{\beta} - \frac{\beta}{\alpha}\right)^2$.

यदि α, β समीकरण $ax^2 + bx + c = 0$ के मूल हो तो $\left(\frac{\alpha}{\beta} - \frac{\beta}{\alpha}\right)^2$ का मान ज्ञात कीजिए।

Ans. $= \frac{(b^2 - 4ac)b^2}{a^2c^2}$

Sol. Equation : $ax^2 + bx + c = 0$

α, β are the roots

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Now $\left(\frac{\alpha}{\beta} - \frac{\beta}{\alpha}\right)^2 = \frac{\alpha^2}{\beta^2} + \frac{\beta^2}{\alpha^2} - 2\frac{\alpha}{\beta}\cdot\frac{\beta}{\alpha}$

$$= \frac{\alpha^4 + \beta^4}{\alpha^2\beta^2} - 2$$

$$= \frac{\left(\alpha^2 + \beta^2\right)^2 - 2\alpha^2\beta^2 - 2\alpha^2\beta^2}{\alpha^2\beta^2}$$

$$= \frac{\left((\alpha + \beta)^2 - 2\alpha\beta\right)^2 - 4\alpha^2\beta^2}{\alpha^2\beta^2}$$

$$= \frac{\left(\frac{b^2}{a^2} - \frac{2c}{a}\right)^2 - \frac{4c^2}{a^2}}{\frac{c^2}{a^2}}$$

$$= \frac{\frac{b^4}{a^4} + \frac{4c^2}{a^2} - \frac{4b^2c}{a^3} - \frac{4c^2}{a^2}}{\frac{c^2}{a^2}} = \frac{b^2}{a^4}\left(b^2 - 4ac\right) \times \frac{a^2}{c^2}$$

$$= \frac{\left(b^2 - 4ac\right)b^2}{a^2c^2}$$

20. Find the value of $x^3 + x^2 - x + 22$ when $x = 1 + 2i$.

यदि $x = 1 + 2i$. हो तो $x^3 + x^2 - x + 22$ का मान ज्ञात कीजिए।

Ans. 7

Sol. $x = 1 + 2i$
$x - 1 = 2i$
$(x - 1)^2 = -4$
$x^2 + 1 - 2x = -4$
$x^2 - 2x + 5 = 0$
Now dividing $x^3 + x^2 - x + 22$ by $x^2 - 2x + 5$, we get $x + 3$ as quotient and 7 as remainder
So that $x^3 + x^2 - x + 22 = (x + 3)\cdot(x^2 - 2x + 5) + 7$
$= (x + 3) \times 0 + 7 = 7$

21. Find the value of $x^3 - 3x^2 - 8x + 15$ when $x = 3 + i$

यदि $x = 3 + i$ हो तो $x^3 - 3x^2 - 8x + 15$ का मान ज्ञात कीजिए।

Ans. – 15

Sol. $x = 3 + i$
$x - 3 = i$
$x^2 + 9 - 6x = -1$
$x^2 - 6x + 10 = 0$
Now dividing $x^3 - 3x^2 - 8x + 15$ by $x^2 - 6x + 10$, we get $x + 3$ as quotient and -15 as remainder
So that $x^3 - 3x^2 - 8x + 15 = (x + 3)\cdot(x^2 - 6x + 10) - 15$
$= (x + 3) \times 0 - 15$
$= -15$

22. Find the value of $x^3 - ax^2 + 2a^2x + 4a^3$ when $\frac{x}{a} = 1 - \sqrt{-3}$

यदि $\frac{x}{a} = 1 - \sqrt{-3}$ हो तो $x^3 - ax^2 + 2a^2x + 4a^3$ का मान ज्ञात कीजिए।

Ans. zero

Sol. $\frac{x}{a} = 1 - \sqrt{-3}$

$\frac{x}{a} - 1 = -\sqrt{3}i$

$x - a = -\sqrt{3}ai$

$(x - a)^2 = -3a^2$

$x^2 + a^2 - 2ax + 3a^2 = 0$

$x^2 - 2ax + 4a^2 = 0$

Now dividing $x^3 - ax^2 + 2a^2x + 4a^3$ by $x^2 - 2ax + 4a^2$, we get $x + a$ as quotient and 0 as remainder.

So that $x^3 - ax^2 + 2a^2x + 4a^3 = (x + a)\cdot(x^2 - 2ax + 4a^2) + 0$

$= (x + a) \times 0 = 0$

23. If α and β are the roots of $x^2 + px + q = 0$ form the equation whose roots are $(\alpha - \beta)^2$ and $(\alpha + \beta)^2$.

यदि α तथा β समीकरण $x^2 + px + q = 0$ के मूल हो तो वह समीकरण ज्ञात कीजिए जिसके मूल $(\alpha - \beta)^2$ तथा $(\alpha + \beta)^2$ है।

Ans. $x^2 - 2x(p^2 - 2q) + p^2(p^2 - 4q) = 0$

Sol. Equation $x^2 + px + q = 0$

α, β are the roots

$\alpha + \beta = -p$; $\alpha - \beta = \frac{\sqrt{p^2 - 4q}}{1} = \sqrt{p^2 - 4q}$

$\alpha\beta = q$

Sum of new roots $= (\alpha - \beta)^2 + (\alpha + \beta)^2$

$= \alpha^2 + \beta^2 - 2\alpha\beta + \alpha^2 + \beta^2 + 2\alpha\beta$

$= 2(\alpha^2 + \beta^2) = 2\{(\alpha + \beta)^2 - 2\alpha\beta\}$

$= 2\{p^2 - 2q)\} = (2p^2 - 4q)$

Product of new roots $= (\alpha - \beta)^2\cdot(\alpha + \beta)^2$

$= (p^2 - 4q)\cdot p^2$

Then Quadratic Equation $x^2 - (2p^2 - 4q)x + p^2(p^2 - 4q) = 0$

24. Prove that the roots of $(x - a)(x - b) = h^2$ are always real.

सिद्ध कीजिये कि समीकरण $(x - a)(x - b) = h^2$ के मूल हमेशा वास्तविक होंगे।

Ans.

Sol. Equation $(x-a)(x-b)=h^2$

$x^2-(a+b)x+ab-h^2=0$

Now $D=(-(a+b))^2-4(ab-h^2)$

$=a^2+b^2+2ab-4ab+4h^2$

$=(a-b)^2+(2h)^2$

+ve +ve

$\therefore D \geq 0$

So that roots are always real.

25. If x_1 and x_2 are the roots of $ax^2+bx+c=0$, find the value of

(i) $(ax_1+b)^{-2}+(ax_2+b)^{-2}$ (ii) $(ax_1+b)^{-3}+(ax_2+b)^{-3}$

यदि x_1 तथा x_2 समीकरण $ax^2+bx+c=0$ के मूल हो तो निम्न के मान ज्ञात कीजिए।

(i) $(ax_1+b)^{-2}+(ax_2+b)^{-2}$ (ii) $(ax_1+b)^{-3}+(ax_2+b)^{-3}$

Ans. (i) $\frac{b^2-2ac}{a^2c^2}$ (ii) $\frac{b(b^2-3ac)}{a^3c^3}$

Sol. Equation $ax^2+bc+c=0$

x_1, x_2 are the roots.

$$x_1+x_2=-\frac{b}{a}$$

$$x_1 \cdot x_2=\frac{c}{a}$$

$$ax_1^2+bx_1=-c$$

$$ax_1+b=\frac{c}{x_1}$$

$$ax_2^2+bx_2=-c$$

$$ax_2+b=-\frac{c}{x_2}$$

(i) $(ax_1+b)^{-2}+(ax_2+b)^{-2}$

$$=\frac{1}{(ax_1+b)^2}+\frac{1}{(ax_2+b)^2}$$

$$=\frac{x_1^2}{c^2}+\frac{x_2^2}{c^2}$$

$$=\frac{1}{c^2}\left(x_1^2+x_2^2\right)$$

$$= \frac{1}{c^2}\left\{\left(x_1 + x_2\right)^2 - 2x_1x_2\right\}$$

$$= \frac{1}{c^2}\left\{\frac{b^2}{a^2} - \frac{2c}{a}\right\}$$

$$= \frac{b^2 - 2ac}{a^2c^2}$$

(ii) $(ax_1 + b)^{-3} + (ax_2 + b)^{-3}$

$$= \frac{1}{\left(ax_1 + b\right)^3} + \frac{1}{\left(ax_2 + b\right)^3}$$

$$= \frac{-x_1^3}{c^3} - \frac{x_2^3}{c^3}$$

$$= -\frac{1}{c^3}\left\{\left(x_1 + x_2\right)^3 - 3x_1x_2\left(x_1 + x_2\right)\right\}$$

$$= -\frac{1}{c^3}\left\{\frac{-b^3}{a^3} - \frac{3c}{a}\cdot\left(\frac{-b}{a}\right)\right\}$$

$$= -\frac{1}{c^3}\left\{\frac{-b^3 + 3abc}{a^3}\right\}$$

$$= \frac{b^3 - 3abc}{a^3c^3}$$

26. Find the condition that the one root of $ax^2 + bx + c = 0$ shall be n times of the other.

वह प्रतिबन्ध ज्ञात कीजिये जिसके लिये समीकरण $ax^2 + bx + c = 0$ का एक मूल दूसरे मूल का n गुना हो।

Ans. $b^2n = ac(1 + n)^2$

Sol. Equation $ax^2 + bx + c = 0$
roots are α, $n\alpha$

$$\therefore \alpha + n\alpha = \frac{-b}{a}$$

$$\alpha\left(1 + n\right) = \frac{-b}{a}$$

$$\alpha = \frac{-b}{a\left(1 + n\right)}$$

& $n\alpha^2 = \frac{c}{a}$

$$\frac{nb^2}{a^2(1+n)^2} = \frac{c}{a}$$

$nb^2 = ac\,(1+n)^2$

27. If α and β are the roots of $ax^2 + bx + c = 0$, From the equation whose roots are $\alpha^2 + \beta^2$ and $\alpha^{-2} + \beta^{-2}$.

यदि α तथा β समीकरण $ax^2 + bx + c = 0$ के मूल हो तो वह समीकरण ज्ञात करो जिसके मूल $\alpha^2 + \beta^2$ तथा $\alpha^{-2} + \beta^{-2}$ है।

Ans. $a^2c^2x^2 - (b^2 - 2ac)(a^2 + c^2)x + (b^2 - 2ac)^2 = 0$

Sol. Equation $ax^2 + bx + c = 0$

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Sum of two roots = $\alpha^2 + \beta^2 + \alpha^{-2} + \beta^{-2}$

$$= (\alpha+\beta)^2 - 2\alpha\beta + \frac{1}{\alpha^2} + \frac{1}{\beta^2}$$

$$= (\alpha+\beta)^2 - 2\alpha\beta + \frac{\alpha^2+\beta^2}{\alpha^2\beta^2}$$

$$= (\alpha+\beta)^2 - 2\alpha\beta + \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha^2\beta^2}$$

$$= \frac{b^2}{a^2} - \frac{2c}{a} + \frac{\frac{b^2}{a^2} - \frac{2c}{a}}{\frac{c^2}{a^2}}$$

$$= \left(\frac{b^2 - 2ac}{a^2}\right) + \frac{(b^2 - 2ac)}{c^2}$$

Product of new roots = $(\alpha^2 + \beta^2)\cdot(\alpha^{-2} + \beta^{-2})$

$$= (\alpha^2+\beta^2)\cdot\frac{(\alpha^2+\beta^2)}{\alpha^2\beta^2}$$

$$= \frac{\left(\alpha^2+\beta^2\right)^2}{\alpha^2\beta^2}$$

$$= \frac{\left((\alpha+\beta)^2-2\alpha\beta\right)^2}{\alpha^2\beta^2}$$

$$= \left(\frac{\left(\frac{b^2}{a^2}-\frac{2c}{a}\right)^2}{\frac{c^2}{a^2}} \right) = \frac{\left(b^2-2ac\right)^2}{a^4} \times \frac{a^2}{c^2}$$

$$= \frac{\left(b^2-2ac\right)^2}{a^2c^2}$$

Now Quadratic Equation

$$x^2 - \left\{ \frac{b^2-2ac}{a^2} + \frac{b^2-2ac}{c^2} \right\} x + \frac{\left(b^2-2ac\right)^2}{a^2c^2} = 0$$

$a^2c^2x^2 - (b^2 - 2ac)(a^2 + c^2)x + (b^2 - 2ac)^2 = 0$

28. Form the equation whose roots are square of the sum and of the difference of the roots of $2x^2 + 2(m+n)x + m^2 + n^2 = 0$

वह समीकरण ज्ञात करो जिसके मूल समीकरण $2x^2 + 2(m+n)x + m^2 + n^2 = 0$ के मूलों के योग का तथा अन्तर का वर्ग है।

Ans. $x^2 - 4mnx - (m^2 - n^2)^2 = 0$

Sol. $2x^2 + 2(m + n)x + m^2 + n^2 = 0$

α, β are the roots.

$\alpha + \beta = -(m + n)$

$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$$= (m+n)^2 - 4\frac{\left(m^2+n^2\right)}{2} = m^2 + n^2 + 2mn - 2m^2 - 2n^2$$

$= -m^2 - n^2 + 2mn$

$= -(m - n)^2$

$$\alpha\beta = \frac{\left(m^2+n^2\right)}{2}$$

New roots are $(\alpha + \beta)^2, (\alpha - \beta)^2$

Sum of new roots

$= 2\ (\alpha^2 + \beta^2)$

$= 2\ \{(\alpha + \beta)^2 - 2\alpha\beta\}$

$$= 2\left\{(m+n)^2\ 2\cdot\frac{\left(m^2+n^2\right)}{2}\right\}$$

$= 2\ \{m^2 + n^2 + 2mn - m^2 - n^2\}$

$= 4mn$

Product of new roots $= (\alpha + \beta)^2 \cdot (\alpha - \beta)^2$

$= -(m + n)^2\ (m - n)^2 = -(m^2 - n^2)^2$

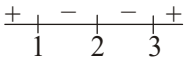
Now Quadratic Equation $x^2 - 4mnx - (m^2 - n^2)^2 = 0$

1. $(x-1)(3-x)(x-2)^2 > 0$

Ans. $(1, 2) \cup (2, 3)$

Sol. $(x-1)(3-x)(x-2)^2 > 0$

$(x-1)(x-3)(x-2)^2 < 0$



$x \in (1, 2) \cup (2, 3)$

2. $\dfrac{6x-5}{4x+1} < 0$

Ans. $(-1/4, 5/6)$

Sol. $\dfrac{6x-5}{4x+1} < 0$

$x \in \left(-\dfrac{1}{4}, \dfrac{5}{6}\right)$

3. $\dfrac{2x-3}{3x-7} > 0$

Ans. $(-\infty, 3/2) \cup (7/3, +\infty)$

Sol. $\dfrac{2x-3}{3x-7} > 0$

$x \in \left(-\infty, \dfrac{3}{2}\right) \cup \left(\dfrac{7}{3}, \infty\right)$

4. $\dfrac{0.5}{x-x^2-1} < 0$

Ans. $(-\infty, +\infty)$

Sol. $\frac{0.5}{x - x^2 - 1} < 0$

$$\frac{1}{2\left(x^2 - x + 1\right)} > 0$$

$$\frac{1}{\left(x^2 - x + 1\right)} > 0$$

for $x^2 - x + 1$

$D = 1 - 4 = -3 < 0$

and $a > 0$

$\therefore x \in R$

5. $\frac{x^2 - 5x + 6}{x^2 + x + 1} < 0$

Ans. (2, 3)

Sol. $\frac{x^2 - 5x + 6}{x^2 + x + 1} < 0$

for $x^2 + x + 1$

$D = b^2 - 4ac$

$= 1 - 4 < 0$

$a > 0$

Now $\therefore x^2 + x + 1 > 0$

for $x^2 - 5x + 6$

$(x - 2)(x - 3) < 0$

$x = 2, 3$

$x \in (2, 3)$

6. $\frac{x^2 + 2x - 3}{x^2 + 1} < 0$

Ans. (–3, 1)

Sol. $\frac{x^2 + 2x - 3}{x^2 + 1} < 0$

for $x^2 + 1 = 0$

$D < 0, a > 0$

$\therefore x^2 + 1 > 0$

Now for $x^2 + 2x - 3 < 0$

$x^2 + 3x - x - 3 < 0$
$x(x + 3) - 1(x + 3) < 0$
$(x + 3)(x - 1) < 0$
$\therefore x \in (-3, 1)$

7. $\dfrac{(x-1)(x+2)^2}{-1-x} < 0$

Ans. $(-\infty, -2) \cup (-2, -1) \cup (1, +\infty)$

Sol. $\dfrac{(x-1)(x+2)^2}{-1-x} < 0$

$\dfrac{(x-1)(x+2)^2}{(x+1)} > 0$

$\therefore x \in (-\infty, -2) \cup (-2, -1) \cup (1, +\infty)$

8. $\dfrac{x^2+4x+4}{2x^2-x-1} < 0$

Ans. $(-1/2, 1)$

Sol. $\dfrac{x^2+4x+4}{2x^2-x-1} < 0$

$\dfrac{(x+2)^2}{2x^2-2x+x-1} < 0$

$\dfrac{(x+2)^2}{2x(x-1)+1(x-1)} < 0$

$\dfrac{(x+2)^2}{(x-1)(2x+1)} < 0$

$x \in \left(-\dfrac{1}{2}, 1\right)$

9. $x^4 - 5x^2 + 4 < 0$

Ans. $(-2, -1) \cup (1, 2)$

Sol. $x^4 - 5x^2 + 4 < 0$
$x^4 - 4x^2 - x^2 + 4 < 0$
$x^2 (x^2 - 4) - 1(x^2 - 4) < 0$
$(x^2 - 4) (x^2 - 1) < 0$
$(x + 2) (x - 2) (x + 1) (x - 1) < 0$
$x \in (-2, -1) \cup (1, 2)$

10. $x^4 - 2x^2 - 63 \leq 0$

Ans. $[-3, 3]$

Sol. $x^4 - 2x^2 - 63 \leq 0$
$x^4 - 9x^2 + 7x^2 - 63 \leq 0$
$x^2 (x^2 - 9) + 7 (x^2 - 9) \leq 0$
$(x^2 - 9) (x^2 + 7) \leq 0$
$(x + 3) (x - 3) (x^2 + 7) \leq 0$
for $x^2 + 7 = 0$
$a > 0, D < 0$
$\therefore x^2 + 7$ always positive.
Now
$x \in [-3, 3]$

11. $\dfrac{3}{x-2} < 1$

Ans. $(-\infty, 2) \cup (5, +\infty)$

Sol. $\dfrac{3}{x-2} < 1$

$\dfrac{3}{x-2} - 1 < 0$

$\dfrac{3 - x + 2}{x-2} < 0$

$\dfrac{(x-5)}{(x-2)} > 0$

$x \in (-\infty, 2) \cup (5, \infty)$

12. $\frac{1}{x-1} \le 2$

Ans. $(-\infty, 1) \cup [3/2, +\infty)$

Sol. $\frac{1}{x-1} \le 2$

$$-\frac{1}{x-1} - 2 \le 0$$

$$\frac{1-2x+2}{x-1} \le 0$$

$$\frac{(2x-3)}{(x-1)} \ge 0$$

$$x \in (-\infty, 1) \cup \left[\frac{3}{2}, +\infty\right)$$

13. $\frac{4x+3}{2x-5} < 6$

Ans. $(-\infty, 5/2) \cup (33/8, +\infty)$

Sol. $\frac{4x+3}{2x-5} < 6$

$$\frac{4x+3}{2x-5} - 6 < 0$$

$$\frac{4x+3-12x+30}{2x-5} < 0$$

$$\frac{33-8x}{2x-5} < 0$$

$$\frac{8x-33}{2x-5} > 0$$

$$x \in \left(-\infty, \frac{5}{2}\right) \cup \left(\frac{33}{8}, +\infty\right)$$

14. $\frac{5x-6}{x+6} < 1$

Ans. $(-6, 3)$

Sol. $\frac{5x-6}{x+6} < 1$

$\frac{5x-6-x-6}{x+6} < 0$

$\frac{4x-12}{x+6} < 0$

$\frac{x-3}{x+6} < 0$

$x \in (-6, 3)$

15. $\frac{5x+8}{4-x} < 2$

Ans. $(-\infty, 0) \cup (4, +\infty)$

Sol. $\frac{5x+8}{4-x} < 2$

$\frac{5x+8-8+x}{4-x} < 0$

$\frac{6x}{4-x} < 0$

$\frac{x}{x-4} > 0$

$x \in (-\infty, 0) \cup (4, +\infty)$

16. $\frac{x-1}{x+3} > 2$

Ans. $(-7, -3)$

Sol. $\frac{x-1}{x+3} > 2$

$\frac{x-1-2x-6}{x+3} > 0$

$\frac{-x-7}{x+3} > 0$

$$\frac{x+7}{x+3} < 0$$

$x \in (-7, -3)$

17. $\frac{7x-5}{8x+3} > 4$

Ans. $(-17/25, -3/8)$

Sol. $\frac{7x-5}{8x+3} > 4$

$$\frac{7x-5-32x-12}{8x+3} > 4$$

$$\frac{-25x-17}{8x+3} > 0$$

$$\frac{25x+17}{8x+3} < 0$$

$$x \in \left(\frac{-17}{25}, \frac{-3}{8}\right)$$

18. $\frac{x}{x-5} > \frac{1}{2}$

Ans. $(-\infty, -5) \cup (5, +\infty)$

Sol. $\frac{x}{x-5} > \frac{1}{2}$

$$\frac{2x-x+5}{2(x-5)} > 0$$

$$\frac{x+5}{x-5} > 0$$

$x \in (-\infty, -5) \cup (5, +\infty)$

19. $\frac{5x-1}{x^2+3} < 1$

Ans. $(-\infty, 1) \cup (4, +\infty)$

Sol. $\frac{5x-1}{x^2+3}<1$

$\frac{5x-1}{x^2+3}-1<0$

$\frac{5x-1-x^2-3}{x^2+3}<0$

$\frac{-x^2+5x-4}{x^2+3}<0$

$\frac{x^2-5x+4}{x^2+3}>0$

$\because$ x^2+3 is always true.

Now, for $x^2-5x+4>0$

$x^2-4x-x+4>0$

$x(x-4)-1(x-4)>0$

$(x-4)(x-1)>0$

$x \in (-\infty, 1) \cup (4, +\infty)$

20. $\frac{x-2}{x^2+1}<-\frac{1}{2}$

Ans. $(-3, 1)$

Sol. $\frac{x-2}{x^2+1}<\frac{-1}{2}$

$\frac{x-2}{x^2+1}+\frac{1}{2}<0$

$\frac{2x-4+x^2+1}{2\left(x^2+1\right)}<0$

$\frac{x^2+2x-3}{\left(x^2+1\right)}<0$

$\because$ x^2+1 is always +ve

$x^2+2x-3<0$

$x^2+3x-x-3<0$

$x(x+3)-1(x+3)<0$

$(x+3)(x-1)<0$

$x \in (-3, 1)$

21. $\frac{x+1}{(x-1)^2} < 1$

Ans. $(-\infty, 0) \cup (3, +\infty)$

Sol. $\frac{x+1}{(x-1)^2} - 1 < 0$

$\frac{x+1-x^2-1+2x}{(x-1)^2} < 0$

$\frac{-x^2+3x}{(x-1)^2} < 0$

$\frac{-x(x-3)}{(x-1)^2} < 0$

$\frac{x(x-3)}{(x-1)^2} > 0$

$x \in (-\infty, 0) \cup (3, +\infty)$

22. $\frac{x^2-7x+12}{2x^2+4x+5} > 0$

Ans. $(-\infty, 3) \cup (4, +\infty)$

Sol. $\frac{x^2-7x+12}{2x^2+4x+5} > 0$

for $2x^2 + 4x + 5$

$D = 16 - 40$

$= -24 < 0$

$\therefore$ $2x^2 + 4x + 5$ always +ve.

Now for

$x^2 - 7x + 12 > 0$

$x^2 - 4x - 3x + 12 > 0$

$x(x-4) - 3(x-4) > 0$

$(x-4)(x-3) > 0$

$x \in (-\infty, 3) \cup (4, \infty)$

23. $\frac{x^2+6x-7}{x^2+1} \le 2$

Ans. $(-\infty, +\infty)$

Sol. $\frac{x^2+6x-7}{x^2+1} \le 2$

$x^2 + 6x - 7 \le 2x^2 + 2$

$x^2 + 6x - 7 - 2x^2 - 2 \le 0$

$-x^2 + 6x - 9 \le 0$

$x^2 - 6x + 9 \ge 0$

$(x - 3)^2 \ge 0$

$x \in R$

24. $\frac{x^4+x^2+1}{x^2-4x-5} < 0$

Ans. $(-1, 5)$

Sol. $\frac{x^4+x^2+1}{x^2-4x-5} < 0$

for $x^4 + x^2 + 1$

$D < 0$, $\therefore$ $x^2 + x^2 + 1$ always +ve

Now, for $x^2 - 4x - 5 < 0$

$x^2 - 5x + x - 5 < 0$

$x(x - 5) + 1(x - 5) < 0$

$(x - 5)(x + 1) < 0$

$x \in (-1, 5)$

25. $\frac{1+3x^2}{2x^2-21x+40} < 0$

Ans. $(5/2, 8)$

Sol. $\frac{1+3x^2}{2x^2-21x+40} < 0$

$\because$ $1 + 3x^2$ always positive

Now, for $2x^2 - 21x + 40 < 0$

$2x^2 - 16x - 5x + 40 < 0$

$2x(x - 8) - 5(x - 8) < 0$

$(x-8)(2x-5)<0$

$x \in \left(\frac{5}{2}, 8\right)$

26. $\frac{1+x^2}{x^2-5x+6}<0$

Ans. $(2, 3)$

Sol. $\frac{1+x^2}{x^2-5x+6}<0$

$\because$ $1+x^2$ always positive

Now, for $x^2-5x+6<0$

$x^2-3x-2x+6<0$

$x(x-3)-2(x-3)<0$

$(x-2)(x-3)<0$

$x \in (2, 3)$

27. $\frac{x^4+x^2+1}{x^2-4x-5}>0$

Ans. $(-\infty, -1) \cup (5, +\infty)$

Sol. $\frac{x^4+x^2+1}{x^2-4x-5}>0$

$\because$ x^4+x^2+1 always positive.

Now, for $x^2-4x-5>0$

$x^2-5x+x-5>0$

$x(x-5)+1(x-5)>0$

$(x-5)(x+1)>0$

$x \in (-\infty, -1) \cup (5, \infty)$

28. $\frac{1-2x-3x^2}{3x-x^2-5}>0$

Ans. $(-\infty, -1) \cup (1/3, +\infty)$

Sol. $\frac{1-2x-3x^2}{3x-x^2-5}>0$

$$\frac{3x^2+2x-1}{x^2-3x+5}>0$$

$$\frac{3x^2+3x-x-1}{x^2-3x+5}>0$$

$$\frac{3x(x+1)-1(x+1)}{x^2-3x+5}>0$$

$$\frac{(3x-1)(x+1)}{x^2-3x+5}>0$$

for $x^2 - 3x + 5$

$a > 0, D < 0$ always positive

Now, for

$(3x - 1)(x + 1) > 0$

$$x\in(-\infty,1)\cup\left(\frac{1}{2},\infty\right)$$

29. $$\frac{x^2-5x+7}{-2x^2+3x+2}>0$$

Ans. $(-1/2, 2)$

Sol. $$\frac{x^2-5x+7}{-2x^2+3x+2}>0$$

$$\frac{x^2-5x+7}{2x^2-3x-2}<0$$

for $x^2 - 5x + 7$

$a > 0, D = 25 - 28 = -3 < 0$

$\therefore x^2 - 5x + 7 = +ve$

Now, for $2x^2 - 3x - 2 < 0$

$2x^2 - 4x + x - 2 < 0$

$2x(x - 2) + 1(x - 2) < 0$

$(x - 2)(2x + 1) < 0$

$$x\in\left(\frac{-1}{2},2\right)$$

30. $\frac{2x^2-3x-459}{x^2+1}>1$

Ans. $(-\infty,-20)\cup(23,+\infty)$

Sol. $\frac{2x^2-3x-459}{x^2+1}>1$

$2x^2-3x-459-x^2-1>0$

$x^2-3x-460>0$

$x^2-23x+20x-460>0$

$x(x-23)+20(x-23)>0$

$(x-23)(x+20)>0$

$x\in(-\infty,-20)\cup(23,+\infty)$

31. $\frac{x^2-1}{x^2+x+1}<1$

Ans. $(-2,+\infty)$

Sol. $\frac{x^2-1}{x^2+x+1}<1$

$x^2-1-x^2-x-1<0$

$-x-2<0$

$x+2>0$

$x>-2$

$x\in(-2,+\infty)$

32. $\frac{x}{x^2-3x-4}>0$

Ans. $(-1,0)\cup(4,+\infty)$

Sol. $\frac{x}{x^2-3x-4}>0$

$\frac{x}{x^2-4x+x-4}>0$

$\frac{x}{x(x-4)+1(x-4)}>0$

$$\frac{x}{(x+1)(x-4)} > 0$$

$$x \in (-1,0) \cup (4,+\infty)$$

33. $\dfrac{x^2+7x+10}{x+2/3} > 0$

Ans. $(-5,-2) \cup (-2/3, +\infty)$

Sol. $$\frac{x^2+7x+10}{x+\frac{2}{3}} > 0$$

$$\frac{(x+2)(x+5)}{\left(x+\frac{2}{3}\right)} > 0$$

$$x \in (-5,-2) \cup \left(-\frac{2}{3}, \infty\right)$$

34. $\dfrac{3x^2-4x-6}{2x-5} < 0$

Ans. $(-\infty, 2-\sqrt{22}/3) \cup ((2+\sqrt{22})/3, 5/2)$

Sol. $$\frac{3x^2-4x-6}{2x-5} < 0$$

for $3x^2 - 4x - 6 = 0$

$$x = \frac{2+\sqrt{22}}{3}, \frac{2-\sqrt{22}}{3}$$

Now,

$$\frac{\left(x-\frac{(2+\sqrt{22})}{3}\right)\left(x-\frac{(2-\sqrt{22})}{3}\right)}{(2x-5)} < 0$$

$(-\infty, 2-\sqrt{22}/3) \cup ((2+\sqrt{22})/3, 5/2)$

35. $\dfrac{17-15x-2x^2}{x+3}<0$

Ans. $(-17/2,-3)\cup(1,+\infty)$

Sol. $\dfrac{17-15x-2x^2}{x+3}<0$

$$\frac{2x^2+15x-17}{(x+3)}>0$$

$$\frac{2x^2-2x+17x-17}{(x+3)}>0$$

$$\frac{2x(x-1)+17(x-1)}{(x+3)}>0$$

$$\frac{(x-1)(2x+17)}{(x+3)}>0$$

$$x\in\left(\frac{-17}{2},-3\right)\cup(1,+\infty)$$

36. $\dfrac{x^2-9}{3x-x^2-24}<0$

Ans. $(-\infty,-3)\cup(3,+\infty)$

Sol. $\dfrac{x^2-9}{3x-x^2-24}<0$

$$\frac{(x+3)(x-3)}{x^2-3x+24}>0$$

for $x^2-3x+24$

$a>0, D<0$

$\therefore$ $x^2-3x+24$ always +ve

Now, for $(x+3)(x-3)>0$

$$x\in(-\infty,-3)\cup(3,+\infty)$$

37. $\frac{x+7}{x-5}+\frac{3x+1}{2}\geq 0$

Ans. $[1, 3] \cup (5, +\infty)$

Sol. $\frac{x+7}{x-5}+\frac{3x+1}{2}\geq 0$

$$\frac{2x+14+(x-5)(3x+1)}{2(x-5)}\geq 0$$

$$\frac{2x+14+3x^2+x-15x-5}{(x-5)}\geq 0$$

$$\frac{3x^2-13x+x+9}{(x-5)}\geq 0$$

$$\frac{3x^2-12x+9}{(x-5)}\geq 0$$

$$\frac{x^2-4x+3}{x-5}\geq 0$$

$$\frac{(x-1)(x-3)}{(x-5)}\geq 0$$

$$x\in[1,3]\cup(5,+\infty)$$

38. $2x^2+\frac{1}{x}>0$

Ans. $(-\infty, -1/\sqrt[3]{2}) \cup (0, +\infty)$

Sol. $2x^2+\frac{1}{x}>0$

$$\frac{2x^3+1}{x}>0$$

$$x=0,\left(\frac{-1}{2}\right)^{\frac{1}{3}}$$

$$x \in \left(-\infty, \left(-\frac{1}{2} \right)^{\frac{1}{3}} \right) \cup (0, +\infty)$$

39. $\dfrac{x^2 - x - 6}{x^2 + 6x} \geq 0$

Ans. $(-\infty, -6) \cup [-2, 0) \cup [3, +\infty)$

Sol. $\dfrac{x^2 - x - 6}{x^2 + 6x} \geq 0$

$$\frac{x^2 - 3x + 2x - 6}{x(x+6)} \geq 0$$

$$\frac{x(x-3) + 2(x-3)}{x \cdot (x+6)} \geq 0$$

$$\frac{(x+2)(x-3)}{x(x+6)} \geq 0$$

$$x \in (-\infty, -6) \cup [-2, 0) \cup [3, +\infty)$$

40. $\dfrac{x^2 - 5x + 6}{x^2 - 11x + 30} < 0$

Ans. $(2, 3) \cup (5, 6)$

Sol. $\dfrac{x^2 - 5x + 6}{x^2 - 11x + 30} < 0$

$$\frac{(x-2)(x-3)}{(x-5)(x-6)} < 0$$

$$x \in (2, 3) \cup (5, 6)$$

1. The roots of the equation $(b + c)x^2 - (a + b + c)\,x + a = 0$ $(a, b, c \in Q, b + c \neq a)$ are :

(A) irrational and different (B) rational and different

(C) imaginary and different (D) real and equal

समीकरण $(b + c)x^2 - (a + b + c)\,x + a = 0$ $(a, b, c \in Q, b + c \neq a)$ के मूल है :

(A) अपरिमेय एवं असमान (B) परिमेय एवं असमान

(C) काल्पनिक एवं असमान (D) वास्तविक एवं समान

Ans. B

Sol. Equation $(b + c)x^2 - (a + b + c)\,x + a = 0$ $(a, b, c \in Q, b + c \neq a)$

$D = (a + b + c)^2 - 4a\,(b + c)$

$D = \{a + (b + c)\}^2 - 4a\,(b + c)$

$D = \{a - (b + c)\}^2 =$ perfect square

$\therefore$ roots are rational and different

2. The imaginary roots of the equation $(x^2 + 2)^2 + 8x^2 = 6x\,(x^2 + 2)$ are :

(A) $1 \pm i$ (B) $2 \pm i$ (C) $-1 \pm i$ (D) None of these

समीकरण $(x^2 + 2)^2 + 8x^2 = 6x\,(x^2 + 2)$ के काल्पनिक मूल हैं :

(A) $1 \pm i$ (B) $2 \pm i$ (C) $-1 \pm i$ (D) इनमें से कोई नहीं

Ans. A

Sol. $(x^2 + 2)^2 + 8x^2 = 6x\,(x^2 + 2)$

$x^4 + 4 + 4x^2 + 8x^2 = 6x^3 + 12x$

$x^4 - 6x^3 + 12x^2 - 12x + 4 = 0$

divided by x^2

$$x^2 - 6x + 12 - \frac{12}{x} + \frac{4}{x^2} = 0$$

$$\left(x^2 + \frac{4}{x^2}\right) - 6\left(x + \frac{2}{x}\right) + 12 = 0$$

put $x + \frac{2}{x} = t$

$$\Rightarrow \left(x^2 + \frac{4}{x^2}\right) = t^2 - 4$$

$t^2 - 4 - 6t + 12 = 0$

$t^2 - 6t + 8 = 0$

$t^2 - 4t - 2t + 8 = 0$

$t(t - 4) - 2\,(t - 4) = 0$

$(t-4)(t-2)=0$

$t=4, 2$

for $t=4$

$$x+\frac{2}{x}=4$$

$x^2+2=4x$

$x^2-4x+2=0$

$D>0$

for $t=2$

$$x+\frac{2}{x}=2$$

$x^2-2x+2=0$

$\because\ D<0$

$$\therefore x=\frac{2\pm\sqrt{4-8}}{2}$$

$$x=\frac{2\pm\sqrt{-4}}{2}$$

$$x=\frac{2\pm 2i}{2}$$

$x=1\pm i$

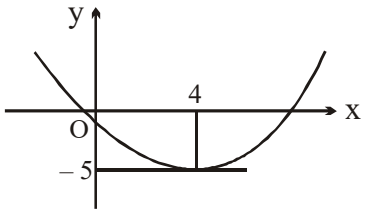
3. The graph of the quadratic trinomial $y=ax^2+bx+c$ has its vertex at $(4,-5)$ and two x-intercepts one positive and one negative. Which of the following hold good ?

द्विघात बहुपद $y=ax^2+bx+c$ के वक्र का शीर्ष $(4,-5)$ तथा x-अन्तःखण्डो में से एक धनात्मक तथा ऋणात्मक है। निम्न में से कौन सत्य है?

(A) $a>0$ (B) $b<0$ (C) $c<0$ (D) $8a=b$

Ans. ABC

Sol.



from figure $a>0 \Rightarrow$ (A) and $-\frac{b}{2a}=4 \Rightarrow -\frac{b}{2a}>0;\ \therefore\ b<0 \Rightarrow$ (B)

$f(0)=c<0 \Rightarrow$ (C)

Also $-\frac{b}{2a}=4 \Rightarrow 8a+b=0 \Rightarrow$ (D) is incorrect

4. The roots of $a_1 x^2 + b_1 x + c_1 = 0$ are reciprocal of the roots of the equation $a_2x^2 + b_2x + c_2 = 0$, if :

समीकरण $a_1 x^2 + b_1 x + c_1 = 0$ के मूल, समीकरण $a_2x^2 + b_2x + c_2 = 0$ के मूलों के व्युत्क्रम होंगे, यदि :

(A) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ (B) $\frac{b_1}{b_2} = \frac{c_1}{a_2} = \frac{a_1}{c_2}$ (C) $\frac{a_1}{a_2} = \frac{b_1}{c_2} = \frac{c_1}{b_2}$ (D) $a_1 = \frac{1}{a_2}, b_1 = \frac{1}{b_2}, c_1 = \frac{1}{c_2}$

Ans. B

Sol. Equation $a_2x^2 + b_2x + c_2 = 0$

α, β are the roots.

$x = \alpha$

Quad. in y

$$y = \frac{1}{\alpha} = \frac{1}{x}$$

put $x = \frac{1}{y}$

$$a_2\left(\frac{1}{y}\right)^2 + \frac{b^2}{y} + c_2 = 0$$

$c_2y^2 + b_2y + a_2 = 0$

$c_2x^2 + b_2x + a_2 = 0$ ...(i)

and $a_1x^2 + b_1x + c_1 = 0$ ...(ii)

$\because$ Both roots are equal

$$\therefore \frac{c_2}{a_1} = \frac{b_2}{b_1} = \frac{a_2}{c_1} \quad \text{OR}$$

$$\frac{b_1}{b_2} = \frac{c_1}{a_2} = \frac{a_1}{c_2}$$

5. If $\alpha \neq \beta$ but $\alpha^2 = 5\alpha - 3$, $\beta^2 = 5\beta - 3$, then the equation whose roots are α/β and β/α is :

(A) $x^2 - 5x - 3 = 0$ (B) $3x^2 + 12x + 3 = 0$ (C) $3x^2 - 19x + 3 = 0$ (D) None of these

यदि $\alpha \neq \beta$ किन्तु $\alpha^2 = 5\alpha - 3$, $\beta^2 = 5\beta - 3$, तब वह समीकरण जिसके मूल α/β तथा β/α है, होगी :

(A) $x^2 - 5x - 3 = 0$ (B) $3x^2 + 12x + 3 = 0$ (C) $3x^2 - 19x + 3 = 0$ (D) इनमें से कोई नहीं

Ans. C

Sol. given $\alpha^2 = 5\alpha - 3$ and $\beta^2 = 5\beta - 3$ $\Rightarrow$ α and β are the roots of $x^2 - 5x + 3 = 0$

$$\therefore \quad \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{25 - 6}{3} = \frac{19}{3} \text{ and } \frac{\alpha}{\beta} \cdot \frac{\beta}{\alpha} = 1$$

$\therefore$ equation have $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ as its roots is $3x^2 - 19x + 3 = 0$

6. The number of values of k for which the system of equations

$(k + 1)x + 8y = 4k$; $kx + (k + 3)y = 3k - 1$

has infinitely many solutions is :

(A) 0 (B) 1 (C) 2 (D) infinite

k के मानों की संख्या तो समीकरणों के निकाय

$(k + 1)x + 8y = 4k$; $kx + (k + 3)y = 3k - 1$

के लिये अनन्त हल दें :

(A) 0 (B) 1 (C) 2 (D) अनन्त

Ans. B

Sol. $\frac{k+1}{k} = \frac{8}{k+3} = \frac{4k}{3k-1}$

$k^2 + 4k + 3 = 8k$ and $24k - 8 = 4k^2 + 12k$

$(k - 3)(k - 1) = 0$ $(k - 2)(k - 1) = 0$

$k = 1, 3$ $k = 2, 1$

$3k^2 + 3k - k - 1 = 4k^2$

$(k - 1)^2 = 0$

$k = 1$ Ans.

7. Two students solve a quadratic equation $x^2 + bx + c = 0$. One student solves the equation by taking wrong value of b and gets the roots as 2 and 5, while second student solves it by taking wrong value of c and gets the roots as – 3 and – 4. The correct roots of the equation are :

(A) – 2, – 5 (B) 2, – 5 (C) 2, 10 (D) None of these

दो विद्यार्थी एक द्विघात समीकरण $x^2 + bx + c = 0$ को हल करते हैं। एक विद्यार्थी b का गलत मान लेकर उसे हल करता है, तो 2 तथा 5 मूल प्राप्त होते हैं। दूसरा विद्यार्थी c का गलत मान लेकर उसे हल करता है तो उसे – 3, – 4 मूल प्राप्त होते हैं। समीकरण के सही मूल होंगे :

(A) – 2, – 5 (B) 2, – 5 (C) 2, 10 (D) इनमें से कोई नहीं

Ans. A

Sol. $x^2 + bx + c = 0$

Product of roots = $c = 2 \times 5 = 10$ {by first student : if b is wrong then c will be correct.}

Sum of roots = $-3 - 4 = -7$ {by second student : if c is wrong then b will be correct.}

$\therefore$ Equation $x^2 + 7x + 10 = 0$

$x^2 + 2x + 5x + 10 = 0$

$x(x+2)+5(x+2)=0$
$(x+2)(x+5)=0$
$x=-2,-5$

8. If the quadratic equations $3x^2+ax+1=0$ and $2x^2+bx+1=0$ have a common root,then the value of the expression $5ab-2a^2-3b^2$ is :

यदि द्विघात समीकरण $3x^2+ax+1=0$ तथा $2x^2+bx+1=0$ का एक मूल उभयनिष्ठ है तो व्यंजक $5ab-2a^2-3b^2$ का मान होगा :

(A) 0 (B) 1 (C) –1 (D) None of these

Ans. B

Sol. $3x^2+ax+1=0$ ...(i) $\times 2$
$2x^2+bx+1=0$ ...(ii) $\times 3$
(i) – (ii)
$(2a-3b)x-1=0$

$$x=\frac{1}{(2a-3b)}$$

from equation (i)

$$\frac{3}{(2a-3b)^2}+\frac{a}{(2a-3b)}+1=0$$

$3+a(2a-3b)+(2a-3b)^2=0$
$3+2a^2-3ab+4a^2+9b^2-12ab=0$
$6a^2+9b^2-15ab=-3$
$3(2a^2+3b^2-5ab)=-3$
$2a^2+3b^2-5ab=-1$
$\Rightarrow 5ab-2a^2-3b^2=1$
Option B is correct.

9. If α, β are roots of the equation $(3x+2)^2+p(3x+2)+q=0$, then roots of $x^2+px+q=0$ are :

यदि समीकरण $(3x+2)^2+p(3x+2)+q=0$ के मूल a, b हों, तो $x^2+px+q=0$ के मूल होंगे :

(A) α, β (B) $3\alpha+2, 3\beta+2$ (C) $\frac{1}{3}(\alpha-2), \frac{1}{3}(\beta-2)$ (D) $\alpha-2, \beta-2$

Ans. B

Sol. $(3x+2)^2+p(3x+2)+q=0$
α, β are the roots.
put $3x+2=t$
$t^2+pt+q=0$
γ, δ are the roots.
$3x+2=\gamma \Rightarrow \gamma=3\alpha+2$

$3x + 2 = \delta \Rightarrow \delta = 3\beta + 2$
Option B is correct.

10. If α, β are roots of the equation $x^2 - 5x + 6 = 0$ then the equation whose roots are $\alpha + 3$ and $\beta + 3$ is :

(A) $x^2 - 11x + 30 = 0$ (B) $(x - 3)^2 - 5(x - 3) + 6 = 0$

(C) Both (A) and (B) (D) None of these

यदि α, β समीकरण $x^2 - 5x + 6 = 0$ के मूल है, तो वह समीकरण जिसके मूल $\alpha + 3$ तथा $\beta + 3$ हैं, होगी :

(A) $x^2 - 11x + 30 = 0$ (B) $(x - 3)^2 - 5(x - 3) + 6 = 0$

(C) (A) तथा (B) दोनों (D) इनमें से कोई नहीं

Ans. C

Sol. $x^2 - 5x + 6 = 0$
α, β are the roots
quad. in y
$y = \alpha + 3$
$y = x + 3$
put $x = y - 3$
$(y - 3)^2 - 5(y - 3) + 6 = 0$
$y^2 + 9 - 6y - 5y + 15 + 6 = 0$
$y^2 - 11y + 30 = 0$

11. If α, β are the root of a quadratic equation $x^2 - 3x + 5 = 0$ then the equation whose roots are $(\alpha^2 - 3\alpha + 7)$ and $(\beta^2 - 3\beta + 7)$ is :

यदि α, β द्विघात समीकरण $x^2 - 3x + 5 = 0$ के मूल हैं, तो वह समीकरण जिसके मूल $(\alpha^2 - 3\alpha + 7)$ एवं $(\beta^2 - 3\beta + 7)$ है, होगी :

(A) $x^2 + 4x + 1 = 0$ (B) $x^2 - 4x + 4 = 0$ (C) $x^2 - 4x - 1 = 0$ (D) $x^2 + 2x + 3 = 0$

Ans. B

Sol. $x^2 - 3x + 5 = 0$
α, β are the roots.
$\alpha^2 - 3\alpha + 5 = 0$
$\beta^2 - 3\beta + 5 = 0$
New roots $= \alpha^2 - 3\alpha + 7$
$= (\alpha^2 - 3\alpha + 5) + 2$
$= 2$
And othe root $= \beta^2 - 3\beta + 7$
$= \beta^2 - 3\beta + 5 + 2$
$= 2$
Then new Quadratic Equation
$x^2 - (\text{sum of roots})x + \text{poduct of roots} = 0$
$x^2 - 4x + 4 = 0$

12. If one root of the equations $x^2 + 2x + 3k = 0$ and $2x^2 + 3x + 5k = 0$ is common then the value of k is :

(A) 1, 2 (B) 0, –1 (C) 1, 3 (D) None of these

यदि समीकरणों $x^2 + 2x + 3k = 0$ व $2x^2 + 3x + 5k = 0$ का एक ही उभयनिष्ठ मूल हों, तो k का मान होगा :

(A) 1, 2 (B) 0, –1 (C) 1, 3 (D) इनमें से कोई नहीं

Ans. B

Sol. $x^2 + 2x + 3k = 0,\ 2x^2 + 3x + 5k = 0$

$\because$ One root of the equation is common

$\therefore\ \alpha^2 + 2\alpha + 3k = 0$

$2\alpha^2 + 3\alpha + 5k = 0$

By cross-multiplication

$$\frac{\alpha^2}{10k-9k} = \frac{\alpha}{6k-5k} = \frac{1}{3-4}$$

$$\frac{\alpha^2}{k} = \frac{\alpha}{k} = \frac{1}{-1}$$

$$\Rightarrow \frac{\alpha}{k} = \frac{1}{1}$$

$\alpha = -k$

$$\Rightarrow \frac{\alpha^2}{k} = \frac{\alpha}{k}$$

$k\alpha^2 = k\alpha$

$k\cdot\alpha^2 - k\alpha = 0$

$k\cdot\alpha(\alpha - 1) = 0$

$k\cdot(-k)(-k - 1) = 0$

$k^2(k + 1) = 0$

$k = 0, -1$

13. If the equations $2x^2 + x + k = 0$ and $x^2 + \frac{x}{2} - 1 = 0$ have 2 common roots then the value of k is :

यदि समीकरण $2x^2 + x + k = 0$ तथा $x^2 + \frac{x}{2} - 1 = 0$ के दो उभयनिष्ठ मूल हों, तो k का मान होगा :

(A) 1 (B) 3 (C) –1 (D) –2

Ans. D

Sol. $2x^2 + x + k = 0$ ...(i)

$$x^2 + \frac{x}{2} - 1 = 0$$

$2x^2 + x - 2 = 0$ ...(ii)

Equation (i) and (ii) have 2 common roots.

$\therefore \frac{2}{2} = \frac{1}{1} = \frac{k}{-2}$

$k = -2$

14. If $x^2 + x - 1 = 0$ and $2x^2 - x + \lambda = 0$ have a common root then :

यदि $x^2 + x - 1 = 0$ तथा $2x^2 - x + \lambda = 0$ उभयनिष्ठ मूल रखती हैं, तो :

(A) $\lambda^2 - 7\lambda + 1 = 0$ (B) $\lambda^2 + 7\lambda - 1 = 0$ (C) $\lambda^2 + 7\lambda + 1 = 0$ (D) $\lambda^2 - 7\lambda - 1 = 0$

Ans. C

Sol. $x^2 + x - 1 = 0, 2x^2 - x + \lambda = 0$ have a common root.

$\therefore \alpha^2 + \alpha - 1 = 0$

$2\alpha^2 - \alpha + \lambda = 0$

$$\frac{\alpha^2}{\lambda - 1} = \frac{\alpha}{-2 - \lambda} = \frac{1}{-1 - 2}$$

$$\Rightarrow \alpha = \frac{2 + \lambda}{3}$$

$$\Rightarrow \frac{\alpha^2}{(\lambda - 1)} = \frac{\alpha}{-(2 + \lambda)}$$

$$\alpha = \frac{(\lambda - 1)}{-(2 + \lambda)}$$

$$\frac{(2 + \lambda)}{3} = \frac{(\lambda - 1)}{-(2 + \lambda)}$$

$$(2 + \lambda)^2 = 3(1 - \lambda)$$

$$4 + \lambda^2 + 4\lambda = 3 - 3\lambda$$

$$\lambda^2 + 7\lambda + 1 = 0$$

15. Sachin and Rahul attempted to solve a quadratic equation. Sachin made a mistake in writing down the constant term and ended up in roots (4, 3). Rahul made a mistake in writing down coefficient of x to get roots (3, 2). The correct roots of equation are :

सचिन तथा राहुल एक द्विघात समीकरण को हल करने का प्रयास करते हैं। सचिन ने अचर पद लिखने में गलती की तथा मूल (4, 3) पाए। राहुल ने x का गुणांक गलत लिखा तथा मूल (3, 2) पाए। समीकरण के सही मूल हैं :

(A) –4, –3 (B) 6, 1 (C) 4, 3 (D) –6, –1

Ans. B

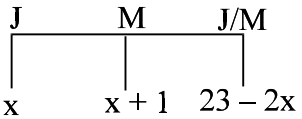
Sol. If c is wrong then sum will be correct.
sum = 4 + 3 = 7
and coefficient of x is wrong then product will be correct
product = 3 × 2 = 6
∴ required equation
$x^2 - 7x + 6 = 0$
$x^2 - 6x - x + 6 = 0$
$(x - 6)(x - 1) = 0$
$x = 1, 6$

16. John has 'x' children by his first wife. Mary has $x + 1$ children by her first husband. They marry and have children of their own. The whole family has 24 children. Assuming that the children of the same parents do not fight, find the maximum possible number of fights that can take place.

जॉन की पहली पत्नी से 'x' बच्चे है। मैरी के पहले पति से $x + 1$ बच्चे है। ये दोनों शादी करते है और इनके भी बच्चे है। पूरे परिवार में 24 बच्चे है। मान लिजिये कि समान अभिभावक के बच्चे लड़ाई नही करते है तो अधिकतम सम्भावित लड़ाईयों की सख्या क्या होगी ?

Ans. 191

Sol. f(x) representing the number of fight functions



$f(x) = x(x + 1) + (x + 1)(23 - 2x) + (23 - 2x)x$
$= (x^2 + x) + (-2x^2 + 21x + 23) - 2x^2 + 23x$
$f(x) = -3x^2 + 45x + 23$
$\because a < 0$

$$\text{Max fights} = -\frac{D}{4a} = \frac{4ac - b^2}{4a} = c - \frac{b^2}{4a}$$

$$= 23 - \frac{45^2}{4 \times (-3)} = 23 + \frac{225}{12}$$

$= 23 + 168.75$
$= 191.75 = 191$

17. Determine the limits between which n must lie in order that the equation $2ax(ax + nc) + (n^2 - 2)c^2 = 0$ may have real roots.

यदि समीकरण $2ax(ax + nc) + (n^2 - 2)c^2 = 0$ के मूल वास्तविक है तो वह सीमा ज्ञात कीजिये जिसमें n स्थित हो।

Ans. –2 and 2

Sol. $2ax(ax + nc) + (n^2 - 2)c^2 = 0$
$2a^2x^2 + 2ancx + (n^2 - 2)c^2 = 0$
$\because D \geq 0$
$4a^2n^2c^2 - 8a^2c^2(n^2 - 2) \geq 0$
$4a^2c^2[n^2 - 2n^2 + 4] \geq 0$
$n^2 - 4 \leq 0$
$(n + 2)(n - 2) \leq 0$
$-2 \leq n \leq 2$
$n \in [-2, 2]$

1. If $x^2 - x - 1 = 0$ then the value of $x^8 + \frac{1}{x^8}$ equals :

यदि $x^2 - x - 1 = 0$ तो $x^8 + \frac{1}{x^8}$ का मान ज्ञात कीजिए :

(A) 21 (B) 33 (C) 42 (D) 47

Ans. D

Sol. If $x^2 - x - 1 = 0$

$\div$ by x

$x - 1 - \frac{1}{x} = 0$

$x - \frac{1}{x} = 1$

$x^2 + \frac{1}{x^2} - 2 = 1$

$x^2 + \frac{1}{x^2} = 3$

$x^4 + \frac{1}{x^4} + 2 = 9$

$x^4 + \frac{1}{x^4} = 7$

$x^8 + \frac{1}{x^8} = 49 - 2$

$x^8 + \frac{1}{x^8} = 47$

2. If x = 1 and x = 2 are solutions of the equation $x^3 + ax^2 + bx + c = 0$ and a + b = 1, then *b* is equal to :

यदि x = 1 तथा x = 2 समीकरण $x^3 + ax^2 + bx + c = 0$ के मूल है तथा a + b = 1 है, तो *b* का मान क्या होगा ?

(A) 2 (B) 4 (C) 5 (D) 3

Ans. C

Sol. If x = 1 and x = 2 are solution of the equation $x^3 + ax^2 + bx + c = 0$ and a + b = 1

at x = 1

$1 + a + b + c = 0$

$a + b + c = -1$ ...(i)

at x = 2

$8 + 4a + 2b + c = 0$

$4a + 2b + c = -8$ ...(ii)

by (ii) – (i)
$3a + b = -7$ ...(iii)
$\because\ a + b = 1$ ...(iv) (Given)
from (iii) – (iv)
$a = -4;\quad b = 5$

3. The sum of the solutions to the equation $\sqrt{3x+4} = 2 + \sqrt{2x-4}$, is :

समीकरण $\sqrt{3x+4} = 2 + \sqrt{2x-4}$, के हलों का योग होगा :

(A) 24 (B) 25 (C) 26 (D) 27

Ans. A

Sol. $\sqrt{3x+4} = 2+\sqrt{2x-4}$

$3x + 4 = 4 + 2x - 4 + 4\sqrt{2x-4}$

$(x + 4) = 4\sqrt{2x-4}$

$x^2 + 16 + 8x = 16(2x - 4)$

$x^2 + 8x - 32x + 16 + 64 = 0$

$x^2 - 24x + 80 = 0$

$\therefore$ Sum of roots = 24

4. If $x^2 + 2xy + 2x + my - 3$ have two rational factors then m is equal to :

यदि $x^2 + 2xy + 2x + my - 3$ के दो परिमेय गुणनखण्ड हों, तो m बराबर है :

(A) 6, 2 (B) – 6, 2 (C) 6, – 2 (D) –6, –2

Ans. C

Sol. $x^2 + 2xy + 2x + my - 3 = 0$

$a = 1, b = 0, h = 1, g = 1, f = \frac{m}{2}, c = -3$

$\because\ D = 0$

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$$0 + 2\times\frac{m}{2}\times1\times1 - 1\times\frac{m^2}{4} - 0 + 3\times1 = 0$$

$$m - \frac{m^2}{4} + 3 = 0$$

$-m^2 + 4m + 12 = 0$

$m^2 - 4m - 12 = 0$

$m^2 - 6m + 2m - 12 = 0$

$m(m - 6) + 2(m - 6) = 0$

$(m - 6)(m + 2) = 0$

$m = -2, 6$

5. Let 'a' be a real number. Number of real roots of the equation $(x^2 + ax + 1)(3x^2 + ax - 3) = 0$, is :

यदि 'a' एक वास्तविक संख्या है। समीकरण $(x^2 + ax + 1)(3x^2 + ax - 3) = 0$ के वास्तविक मूलों की संख्या है :

(A) at least two (B) atmost two (C) exactly two (D) all four

Ans. A

Sol. $x^2 + ax + 1 = 0$, $3x^2 + 8x - 3 = 0$

$D = a^2 - 4$ (roots can be imaginary or equal or distinct) $D = a^2 + 36 > 0$ (two real roots)

$\therefore$ Number of real roots = at least two.

6. If $x = \sqrt{6+\sqrt{6+\sqrt{6+.....}}}$, then :

यदि $x = \sqrt{6+\sqrt{6+\sqrt{6+.....}}}$, तो :

(A) $-2 < x < 3$ (B) $2 < x < 3$ (C) $x = 3$ (D) $x > 3$

Ans. C

Sol. If $x = \sqrt{6+\sqrt{6+\sqrt{6+.....\infty}}}$

$x = \sqrt{6+x}$

$x^2 = 6 + x$

$x^2 - x - 6 = 0$

$x^2 - 3x + 2x - 6 = 0$

$x(x-3) + 2(x-3) = 0$

$(x-3)(x+2) = 0$

$x = 3, x \neq -2$

7. If $a > b > 0$ are two real numbers, the value of,

$\sqrt{ab + (a-b)\sqrt{ab + (a-b)\sqrt{ab + (a-b)\sqrt{ab + ...}}}}$ is :

(A) independent of b (B) independent of a

(C) independent of both a and b (D) dependent on both a and b

यदि $a > b > 0$ दो वास्तविक संख्याऐं हो तो व्यंजक

$\sqrt{ab + (a-b)\sqrt{ab + (a-b)\sqrt{ab + (a-b)\sqrt{ab + ...}}}}$ का मान होगा :

(A) b से स्वतंत्र (B) a से स्वतंत्र

(C) a तथा b दोनों से स्वतंत्र (D) a तथा b दोनों से निर्भर करता है

Ans. A

Sol. We have, $y = \sqrt{ab + (a-b)y}$

$y^2 = ab + (a-b)y$

$\therefore\ y^2 - (a-b)y - ab = 0 \Rightarrow (y-a)(y+b) = 0 \Rightarrow y = a$ or $y = -b$ (not possible) $\Rightarrow$ A

8. If $\alpha, \beta, \gamma, \delta$ are roots of $x^4 - 100x^3 + 2x^2 + 4x + 10 = 0$, then $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}+\frac{1}{\delta}$ is equal to :

यदि $\alpha, \beta, \gamma, \delta$ समीकरण $x^4 - 100x^3 + 2x^2 + 4x + 10 = 0$ के मूल हैं, तो $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}+\frac{1}{\delta}$ बराबर है :

(A) $\frac{2}{5}$ (B) $\frac{1}{10}$ (C) 4 (D) $-\frac{2}{5}$

Ans. D

Sol. $x^4 - 100x^3 + 2x^2 + 4x + 10 = 0$

$\alpha, \beta, \gamma, \delta$ are roots.

$\alpha + \beta + \gamma + \delta = 100$

$\Sigma\alpha\beta = 2$

$\Sigma\alpha\beta\gamma = -4$

$\alpha\beta\gamma\delta = 10$

$\Rightarrow \frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}+\frac{1}{\delta} = \frac{\Sigma\alpha\beta\gamma}{\alpha\beta\gamma\delta} = \frac{-4}{10} = \frac{-2}{5}$

9. If x be real, prove that $\frac{x}{x^2-5x+9}$ must lie between 1 and $\frac{-1}{11}$.

यदि x वास्तविक है तो सिद्ध कीजिये $\frac{x}{x^2-5x+9}$ का मान 1 तथा $\frac{-1}{11}$ के मध्य स्थित होगा।

Ans.

Sol. Let $y = \frac{x}{x^2-5x+9}$

$yx^2 - 5xy + 9y = x$

$yx^2 - (5y+1) + 9y = 0$

$\because D \geq 0, y \neq 0$

$(5y+1)^2 - 36y^2 \geq 0$

$(5y + 1 - 6y)(5y + 1 + 6y) \geq 0$

$-(y-1)(11y+1) \geq 0$

$(y-1)(11y+1) \le 0$

$\frac{-1}{11} \le y \le 1$

$\therefore$ y lie in between 1 and $\frac{-1}{11}$

10. Show that $\frac{x^2-x+1}{x^2+x+1}$ lies between 3 and $\frac{1}{3}$ for all real values of x.

यदि x वास्तविक है तो सिद्ध कीजिये $\frac{x^2-x+1}{x^2+x+1}$ का मान 3 तथा $\frac{1}{3}$ के मध्य स्थित होगा।

Ans.

Sol. Let $y = \frac{x^2-x+1}{x^2+x+1}$

$yx^2 + yx + y - x^2 + x - 1 = 0$

$(y-1)x^2 + (y+1)x + (y-1) = 0$

$y - 1 \neq 0, D \ge 0$

$y \neq 1$

$(y+1)^2 - 4(y-1)^2 \ge 0$

$(y + 1 - 2(y-1))(y + 1 + 2(y-1)) \ge 0$

$(-y+3)(3y-1) \ge 0$

$(y-3)(3y-1) \le 0$

$\because \frac{1}{3} \le y \le 3$

11. If x be real prove that $\frac{x^2+34x-71}{x^2+2x-7}$ can have no value between 5 and 9.

यदि x वास्तविक है तो सिद्ध कीजिये $\frac{x^2+34x-71}{x^2+2x-7}$ का कोई मान 5 तथा 9 के मध्य स्थित नहीं होगा।

Ans.

Sol. Let $y = \frac{x^2+34x-71}{x^2+2x-7}$

$yx^2 + 2xy - 7y = x^2 + 34x - 71$

$(y-1)x^2 + (2y-34)x - 7y + 71 = 0$

$(y-1)x^2 + 2(y-17)x - (7y-71) = 0$
$y \neq 1, D \geq 0$
$4(y-17)^2 + 4(y-1)(7y-71) \geq 0$
$y^2 + 289 - 34y + 7y^2 - 71y - 7y + 71 \geq 0$
$8y^2 - 112y + 360 \geq 0$
$8[y^2 - 14y + 45] \geq 0$
$y^2 - 14y + 45 \geq 0$
$y^2 - 9y - 5y + 45 \geq 0$
$(y-9)(y-5) \geq 0$
$y \in (-\infty, 5) \cup (9, \infty)$
$\therefore$ y do not lie in between 5 and 9.

12. Find the greatest value of $\frac{x+2}{2x^2+3x+6}$ for real value of x.

यदि x वास्तविक है तो $\frac{x+2}{2x^2+3x+6}$ का अधिकतम मान ज्ञात कीजिये।

Ans. $\frac{1}{3}$

Sol. Let $y = \frac{x+2}{2x^2+3x+6}$
$2yx^2 + 3xy + 6y - x - 2 = 0$
$2yx^2 + (3y-1)x + (6y-2) = 0$
$y \neq 0, D \geq 0$
$(3y-1)2 - 4 \cdot 2y(6y-2) \geq 0$
$9y^2 + 1 - 6y - 48y^2 + 16y \geq 0$
$-39y^2 + 10y + 1 \geq 0$
$39y^2 - 10y - 1 \leq 0$
$39y^2 - 13y + 3y - 1 \leq 0$
$13y(3y-1) + 1(3y-1) \leq 0$
$(3y-1)(13y+1) \leq 0$
$\frac{-1}{13} \leq y \leq \frac{1}{3}$
$y_{max} = \frac{1}{3}$

13. For what values of m, will the expression $y^2+2xy+2x+my-3$ be capable of resolution into two rational factors?

m के किन मानों के लिये व्यंजक $y^2 + 2xy + 2x + my - 3$ दो परिमेय गुणनखण्डों में विभाजित हो सकता है ?

Ans. $m = -2$

Sol. $y^2 + 2xy + 2x + my - 3 = 0$

$a = 0, b = 1, h = 1, g = 1, f = \frac{m}{2}, c = -3$

$\because D = 0$

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$0 + 2 \times \frac{m}{2} \times 1 \times 1 - 0 - 1 \times 1 - (-3) \times 1^2 = 0$

$m - 1 + 3 = 0$

$m = -2$

14. Show that the expression $A(x^2 - y^2) - xy(B - C)$ always admits of two real linear factors.

सिद्ध कीजिये व्यंजक $A(x^2 - y^2) - xy(B - C)$ हमेशा दो रेखीय गुणनखण्डों का गुणन है।

Ans.

Sol. $A(x^2 - y^2) - xy(B - C) = 0$

$Ax^2 - Ay^2 - xy(B - C) = 0$

$a = A, b = -A, h = \frac{C - B}{2}, g = 0, f = 0, c = 0$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$= 0 + 0 - 0 - 0$

$= 0$. Hence proved.

1. Given $\frac{x^2+y^2}{x^2-y^2}+\frac{x^2-y^2}{x^2+y^2}=k$ then $\frac{x^8+y^8}{x^8-y^8}$ in terms of k is equal to :

यदि $\frac{x^2+y^2}{x^2-y^2}+\frac{x^2-y^2}{x^2+y^2}=k$ तो $\frac{x^8+y^8}{x^8-y^8}$ का मान k के रूप में होगा :

(A) $\frac{k^2+1}{k^2-1}$ (B) $\frac{k^2+4}{k^2-4}$ (C) $\frac{k^2+4}{4k}$ (D) $\frac{k^2+8}{8k}$

Ans. C

Sol. Given $\frac{x^2+y^2}{x^2-y^2}+\frac{x^2-y^2}{x^2+y^2}=k$

$$\frac{\left(x^2+y^2\right)^2+\left(x^2-y^2\right)^2}{x^4-y^4}=k$$

$$\frac{x^4+y^4+2x^2y^2+x^4+y^4-2x^2y^2}{x^4-y^4}=k$$

$$\frac{2\left(x^4+y^4\right)}{\left(x^4-y^4\right)}=k$$

$$\frac{x^4+y^4}{x^4-y^4}=\frac{k}{2}$$

C & D Rule

$$\frac{2x^4}{2y^4}=\frac{k+2}{k-2}$$

Squaring both sides

$$\frac{x^8}{y^8}=\frac{k^2+4+4k}{k^2+4-4k}$$

again C & D rule

$$\frac{x^8+y^8}{x^8-y^8}=\frac{k^2+4+4k+k^2+4-4k}{k^2+4+4k-k^2-4+4k}$$

$$=\frac{2\left(k^2+4\right)}{8k}=\frac{k^2+4}{4k}$$

2. If one root of the equation $x^2-(p+1)x+p^2+p-8=0$ is greater than 2 and the other root is smaller than 2, then p is such that :

(A) $-\frac{1}{3}<p<3$ (B) $-2<p<3$ (C) $2<p<3$ (D) None of these

यदि समीकरण $x^2-(p+1)x+p^2+p-8=0$ का एक मूल 2 से अधिक तथा दूसरा 2 से कम हो, तो p इस प्रकार होगा, कि :

(A) $-\frac{1}{3}<p<3$ (B) $-2<p<3$ (C) $2<p<3$ (D) इनमें से कोई नहीं

Ans. B

Sol. $f(2)=4-2(p+1)+p^2+p-8<0$

$\Rightarrow (p-3)(p+2)<0 \Rightarrow -2<p<3$

3. If both the roots of the equation $x^2-9x+a=0$ are positive and one is greater than 3 and other is less than 3, then all possible values of a is :

(A) $0<a<18$ (B) $-1<a<2$ (C) $-18<a<0$ (D) None of these

यदि समीकरण $x^2-9x+a=0$ के दोनों मूल धनात्मक हो तथा एक मूल 3 से बड़ा व एक 3 से छोटा है, तो a के सभी संभावित मान होंगे:

(A) $0<a<18$ (B) $-1<a<2$ (C) $-18<a<0$ (D) इनमें से कोई नहीं

Ans. A

Sol. $x^2-9x+a=0$

$f(3)<0$

$9-27+a<0$

$-18+a<0$

$a<18$

$\because$ roots are positive

$\therefore$ product of roots >0

$\frac{a}{1}>0$

$a>0$

$\Rightarrow 0<a<18$

4. The least integral value of a for which the equation $x^2-2(a-1)x+(2a+1)=0$ has both the roots positive is :

a का वह न्यूनतम पूर्णांक मान जिसके लिए समीकरण $x^2 - 2(a-1)x + (2a+1) = 0$ के दोनों मूल धनात्मक हों, होगा :

(A) 3 (B) 4 (C) 1 (D) 5

Ans. B

Sol. $x^2 - 2(a-1)x + (2a+1) = 0$

$\Rightarrow f(0) > 0$

$2a + 1 > 0$

$a > \frac{-1}{2}$



$\Rightarrow \frac{-b}{2a} > 0$

$\frac{2(a-1)}{2} > 0$

$a > 1$

And $D \geq 0$

$4(a-1)^2 - 4(2a+1) \geq 0$

$a^2 + 1 - 2a - 2a - 1 \geq 0$

$a(a-4) \geq 0$

$a \leq 0, a \geq 4$

Final : $a \geq 4$

Least integral value of $a = 4$

5. If α, β are the roots of the equation $x^2 - 3x + a = 0$, $a \in R$ and $\alpha < 1 < \beta$ then :

(A) $a \in (-\infty, 2)$ (B) $a \in (-\infty, 9/4)$ (C) $a \in (2, 9/4)$ (D) None of these

यदि α, β समीकरण $x^2 - 3x + a = 0$, $a \in R$ के मूल हैं, एवं $\alpha < 1 < \beta$ तो :

(A) $a \in (-\infty, 2)$ (B) $a \in (-\infty, 9/4)$ (C) $a \in (2, 9/4)$ (D) इनमें से कोई नहीं

Ans. A

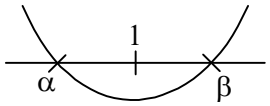
Sol. $x^2 - 3x + a = 0$, $a \in R$ and $\alpha < 1 < \beta$

$f(1) < 0$

$1 - 3 + a < 0$

$a < 2$

$\Rightarrow a \in (-\infty, 2)$



6. If both roots of the equation $x^2 - (m+1)x + (m+4) = 0$ are negative, then m equals :

यदि समीकरण $x^2 - (m+1)x + (m+4) = 0$ के दोनों मूल ऋणात्मक हों, तो m का मान होगा :

(A) $-7 < m < -5$ (B) $-4 < m \leq -3$ (C) $2 < m < 5$ (D) None of these

Ans. B

Sol. $x^2 - (m+1)x + (m+4) = 0$

$\because$ Both roots are negative.

(i) $\therefore D \geq 0$

$(m+1)^2 - 4(m+4) \geq 0$

$m^2 + 1 + 2m - 4m - 16 \geq 0$

$m^2 - 2m - 15 \geq 0$

$m^2 - 5m + 3m - 15 \geq 0$

$m(m-5) + 3(m-5) \geq 0$

$(m-5)(m+3) \geq 0$

$m \leq -3, m \geq 5$



(ii) $\frac{-b}{2a} < 0$

$\frac{(m+1)}{2} < 0$

$m < -1$

(iii) $a \cdot f(0) > 0$

$m + 4 > 0$

$m > -4$

by (i) $\cap$ (ii) $\cap$ (iii)

$m \in (-4, -3]$

7. If α, β are the roots of the equation $2x^2 - (a^3 + 8a - 1)x + (a^3 - 4a) = 0$, then the interval in which 'a' should lie so that $\alpha \in (-\infty, 0)$ and $\beta \in (1, \infty)$ is :

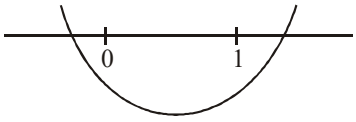
यदि α, β समीकरण $2x^2 - (a^3 + 8a - 1)x + (a^3 - 4a) = 0$ के मूल है तब 'a' का अन्तराल स्थित होना चाहिए जबकि $\alpha \in (-\infty, 0)$ और $\beta \in (1, \infty)$ है :

(A) $(-\infty, -1)$ (B) $(2, \infty)$ (C) $\left(\frac{1}{4}, 2\right)$ (D) $\left(-1, \frac{1}{2}\right)$

Ans. C

Sol. $f(0) < 0 \Rightarrow a^3 - 4a < 0$

$\Rightarrow a(a-2)(a+2) < 0$.....(i)

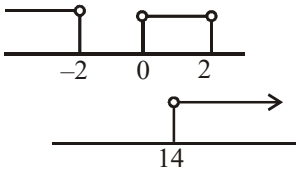


$f(1) < 0 \Rightarrow 2 - a^3 - 8a + 1 + a^3 - 4a < 0$

$-12a + 3 < 0$

$\Rightarrow a > \frac{1}{4}$ ..... (ii)

$a(a-2)(a+2) < 0$ .....(1)



(i) and (ii) gives

$\Rightarrow a \in \left(\frac{1}{4}, 2\right)$

8. If α, β are the roots of the quadratic equation $x^2 - 2p(x-4) - 15 = 0$, then the set of values of 'p' for which one root is less than 1 & the other root is greater than 2 is :

यदि द्विघात समीकरण $x^2 - 2p(x-4) - 15 = 0$ के मूल α, β हो तो इनमें से एक मूल 1 से छोटा और दूसरा मूल 2 से बड़ा होने के लिए 'p' के मानों का समुच्चय है :

(A) $(7/3, \infty)$ (B) $(-\infty, 7/3)$ (C) $x \in R$ (D) $(2, \infty)$

Ans. B

Sol. $x^2 - 2px + (8p - 15) = 0$

$f(1) < 0$ and $f(2) < 0$

$\Rightarrow f(1) = 1 - 2p + 8p - 15 < 0$

$\Rightarrow p < 7/3$

and $f(2) = 4 - 4p + 8p - 15 < 0$

$\Rightarrow 4p - 11 < 0 \Rightarrow p < \frac{11}{4}$

Hence $p \in \left(-\infty, \frac{7}{3}\right)$

9. If the solution set of $|x - k| < 2$ is a subset of the solution set of the inequality $\frac{2x-1}{x+2} < 1$, then the number of possible integral value(s) of 'k' is/are :

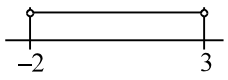
$|x - k| < 2$ का हल यदि $\frac{2x-1}{x+2} < 1$ के हल का उपसमुच्चय है तो 'k' के कुल पूर्णांक हलों की संख्या होगी :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. $\frac{2x-1}{x+2} < 1 \quad \Rightarrow \quad \frac{2x-1}{x+2} - 1 < 0$

or $\frac{2x-1-x-2}{x+2} < 0 \quad \Rightarrow \quad \frac{x-3}{x+2} < 0$; Hence $-2 < x < 3$ ... (1)



also $|x - k| < 2 \quad \Rightarrow \quad -2 < x - k < 2$

$k - 2 < x < k + 2$ ....(2)

hence $k + 2 \le 3 \quad \Rightarrow \quad k \le 1$

and $k - 2 \ge -2 \quad \Rightarrow \quad k \ge 0$

hence $k \in [0, 1] \quad \Rightarrow$ number of integral values of k is $\{0, 1\}$]

10. Find the range of values of 'a' for which the system of equations

$$\left.\begin{aligned} x + ay &= 3 \\ \text{and } ax + 4y &= 6 \end{aligned}\right\} \text{ satisfy } x > 1 \; ; \; y > 0$$

'a' के मानो का परिसर ज्ञात कीजिये जिनके लिये समीकरणो

$$\left.\begin{aligned} x + ay &= 3 \\ \& \quad ax + 4y &= 6 \end{aligned}\right\} \text{ का निकाय } x > 1 \; ; \; y > 0 \text{ को संतुष्ट करता है।}$$

Ans. $(-2, 2) \cup (2, 4)$

Sol. [Hint: $x = \frac{6(a-2)}{a^2-4}, \; y = \frac{3(a-2)}{a^2-4}$]